

IDEA for Teachers: Tutorial

Kolibri, Nextcloud, Wikipedia offline, Files Disk and Backup Disk

Version 1.6, 25 October 2026

Contents

1. What IDEA is 3
   1.1 Words used in this guide 3
   1.2 The apps 3
   1.3 How to use this guide 4
2. Getting connected and opening an app 5
   2.1 Connect to the Wi-Fi 5
   2.2 Open the Console 5
   2.3 First steps in the Console 8
3. Kolibri for teachers 10
   3.1 Signing in 10
   3.2 Create a class 10
   3.3 Enroll learners 11
   3.4 Make a lesson and show it to the class 13
   3.5 Make a quiz 16
   3.6 See progress (Reports) 26
   3.7 Teaching one class period 28
4. Kolibri for learners (try it yourself) 30
   4.1 Do a lesson 30
   4.2 Take a quiz 32
5. Nextcloud for teachers 35
   5.1 Share worksheets with the class (view only) 36
   5.2 Write a document together 37
   5.3 Collect homework (File drop) 39
6. Nextcloud for learners (try it yourself) 42
   6.1 Open class files and co-edit 42
   6.2 Hand in homework 44
7. Wikipedia offline 45
   7.1 Look something up and use it in class 45
8. Your own files: the Files Disk 48
   8.1 Open School Files 48
   8.2 Share a file from School Files with your class 49
9. Keeping data safe: the Backup Disk 51

Appendix: demo accounts 52

1. What IDEA is

IDEA (Initiative for Digital Education in Africa) brings offline computing to rural schools. A small computer in the school, the **IDEA box**, holds ready-made lessons in Kolibri, a place for your own files in Nextcloud, and the whole of Wikipedia. Teachers and learners open them from any classroom computer, laptop or phone on the school Wi-Fi, without internet. This tutorial shows you, step by step, how to use them in your lessons.

> **No internet needed.** Everything in this guide works inside the school. Nothing you do is sent to the internet.

1.1 Words used in this guide

Word	Meaning
IDEA box	The small Raspberry Pi computer in the school that runs the apps. Each box has its own name, on its sticker. The kit's three boxes are **sunny-robin**, **happy-zebra** and **kind-panda**.
Console	The web page that lists the apps and opens them.
Demo apps	**Kolibri Demo**, **Nextcloud Demo** and **Wikipedia (Demo)**: practice copies with example content, on every IDEA box. Use them to follow this tutorial (1.3).
IDEA shortcut	The one shortcut on a classroom computer's desktop. It finds the IDEA box and opens the Console (chapter 2).
System Disk	The IDEA box's own disk. The apps live on it.
IDEA Disk	A USB disk or stick that the Console has prepared. It can become a **Files Disk** or a **Backup Disk**.
Spare disk	An IDEA Disk that is still empty.
Files Disk	An IDEA Disk for your own files. It appears as the folder **School Files** in Nextcloud (chapter 8).
Backup Disk	An IDEA Disk that keeps copies of the apps' data (chapter 9).
Operator	The person at the school who looks after the IDEA box (IDEA for Operators: Tutorial).

1.2 The apps

App	What you use it for	Chapters
Kolibri	Ready-made lessons, videos and exercises (on the kit, for example CK-12, PhET, OpenStax and African Storybook). You make classes, lessons and quizzes and see each learner's progress.	3, 4

App	What you use it for	Chapters
Nextcloud	Your own files: share worksheets with a class, write documents together in **Nextcloud Office**, collect homework.	5, 6, 8
Wikipedia	The encyclopedia, offline: the full English Wikipedia, and a Simple English one in the Demo.	7

1.3 How to use this guide

- This tutorial is for teachers. The school operator has its own: IDEA for Operators: Tutorial.
- Each chapter has numbered steps with screenshots. The screenshots use the demo setup that is on every IDEA box: the demo school **IDEA Demo School** with class **Grade 5** and the demo accounts in the appendix (`admin`, `teacher1`, `teacher2`, `learner1` to `learner3`; teachers and the admin have a password: the account name followed by `911!`; learners sign in in Kolibri with only their username). Your school's names will be different.
- **Practise in the Demo apps.** Every IDEA box of the kit has six apps: the school's own **Kolibri**, **Nextcloud** and **Wikipedia**, and a Demo copy of each. Follow this tutorial in **Kolibri Demo**, **Nextcloud Demo** and **Wikipedia (Demo)**: they hold the demo school, the demo accounts and the example content. The school's own apps stay clean for real lessons.

2. Getting connected and opening an app

All apps open from one web page: the **Console**. How you get to the Console depends on the computer you use.

2.1 Connect to the Wi-Fi

The classroom computers are already connected. Do this only on your own laptop, tablet or phone.

The IDEA router at your school makes two Wi-Fi networks. Their names and first passwords are the same at every IDEA school:

Network	Password (until your school changes it)	Who joins it	What it reaches
IDEA	`idea911!`	Students, and the classroom computers	Only the IDEA box: the Console and its apps. No internet.
IDEA-Teachers	`idea-teachers911!`	Teachers only	The IDEA box, and the internet. Internet is **on**; your IDEA contact can switch it off when the head teacher asks.

- Give students only the **IDEA** password. Keep the **IDEA-Teachers** password among the teachers.
- The classroom computers stay on **IDEA**. For a teacher's classroom computer on **IDEA-Teachers**, ask your IDEA contact.

To connect:

1. Open the Wi-Fi settings on your device.
2. Choose **IDEA-Teachers** (teachers) or **IDEA** (students).
3. Type the password and join.

2.2 Open the Console

There are two ways to open the Console. Use the one that fits the device in front of you.

2.2.1 On a classroom computer

The classroom computers have **one** shortcut on the desktop, named **IDEA**. There are no separate icons for Kolibri, Nextcloud or the other apps.

1. Double-click **IDEA** on the desktop. You can also find it in the app menu. It opens in Chrome.

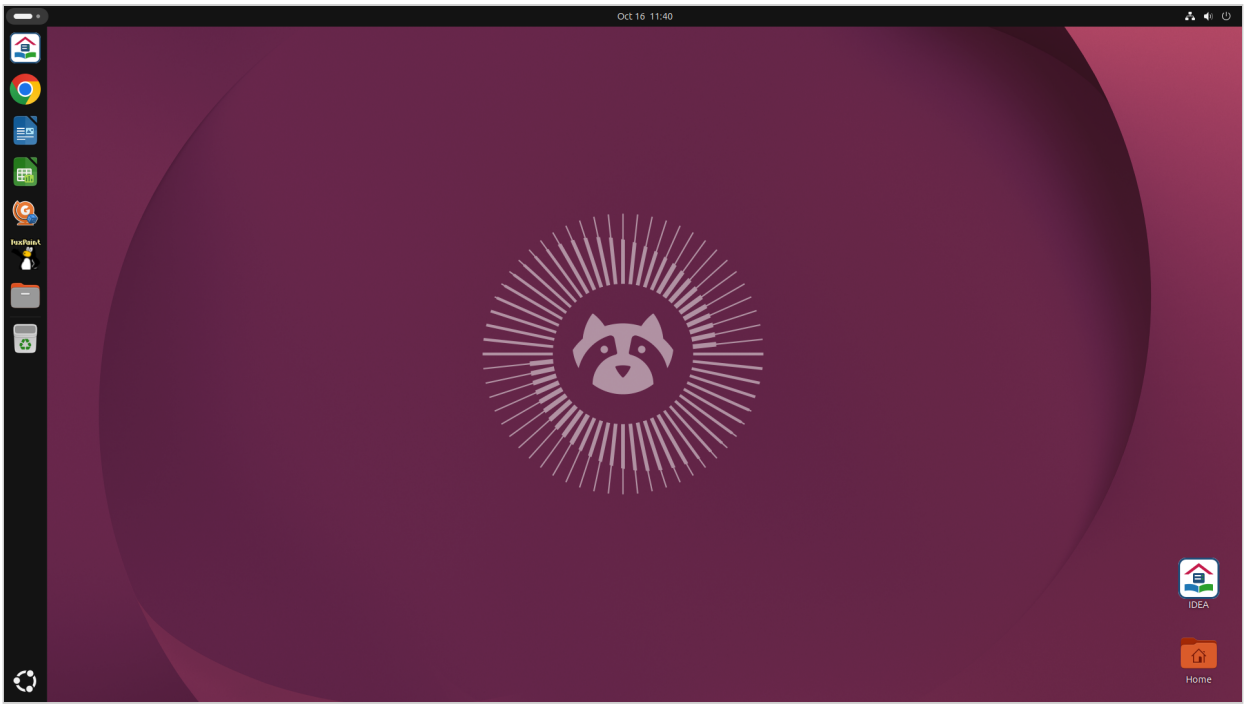


Figure 1. The classroom computer desktop with the one IDEA shortcut

2. A start page titled **IDEA** appears for a moment and says **Looking for the school server…**.

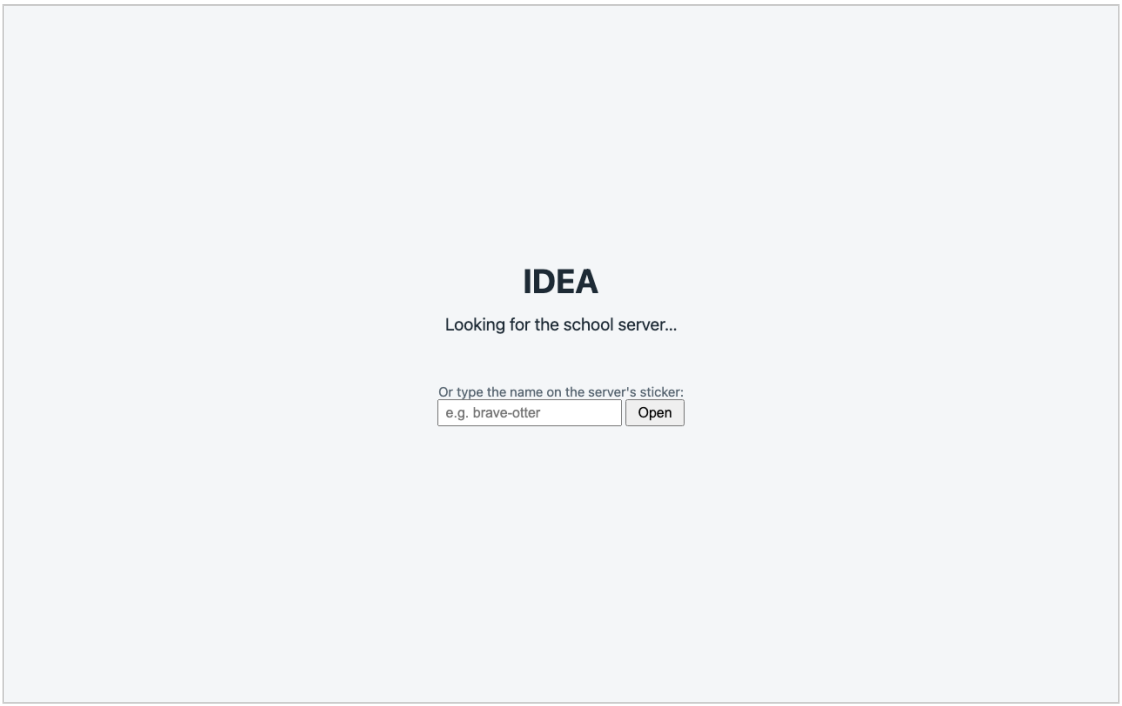


Figure 2. The IDEA start page: Looking for the school server…

3. The Console opens by itself.

If the start page says **No school server was found**, check that the computer is connected to the school network. Then type the first line of the box's sticker, for example `brave-otter`, in the field **Or type the name on the server's sticker**. Use small letters, with `-` between the words. Press **Open**. The start page remembers that IDEA box.

2.2.2 On your own laptop, tablet or phone

1. Connect to the Wi-Fi (2.1).
2. Open the browser.
3. Type `http://idea.local` in the address bar and press Enter.
4. The Console opens.

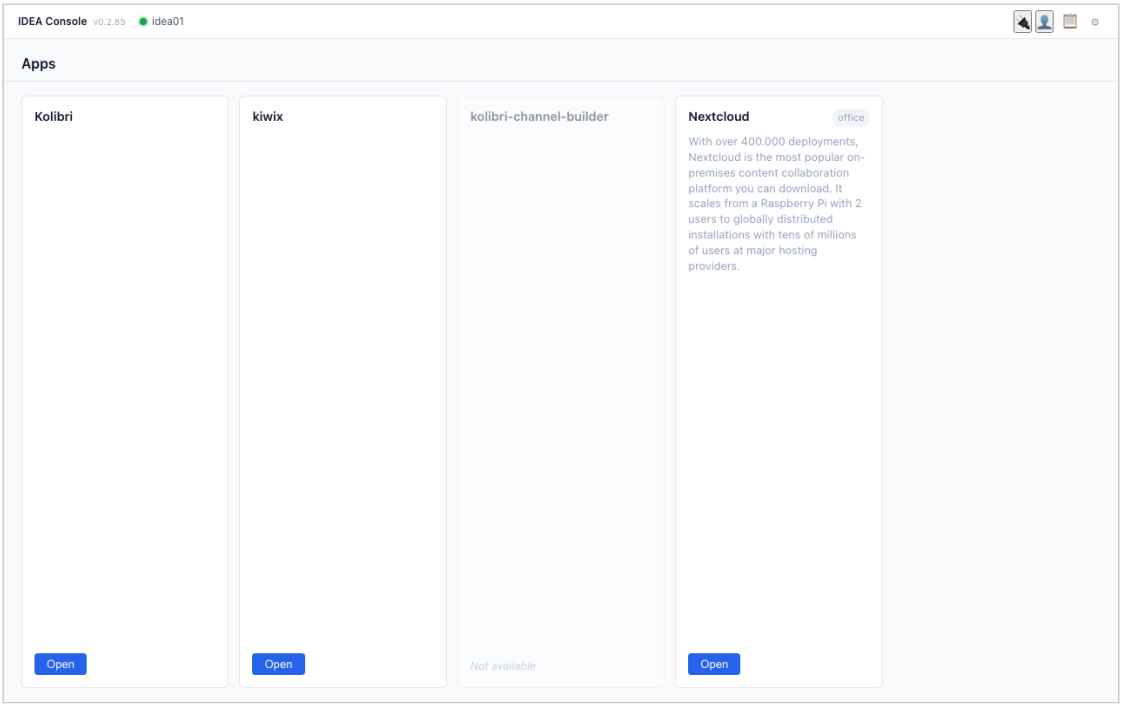


Figure 3. The Console in the browser

5. Bookmark it, so you can open it again with one click. In Chrome, press **Ctrl+D** (on a Mac ⌘+**D**), then **Done**.

If the Console does not open, type the address on the second line of the box's sticker instead, for example `http://brave-otter.local`.

> **Two IDEA boxes at your school?** The Console shows the apps of all of them. Each card says on which box the app runs, for example **on kind-panda**.

2.3 First steps in the Console

1. The Console opens on the **Apps** page: one card per app, each with a short description and a blue **Open** button. The green dot at the top shows the Console is connected to the IDEA box.

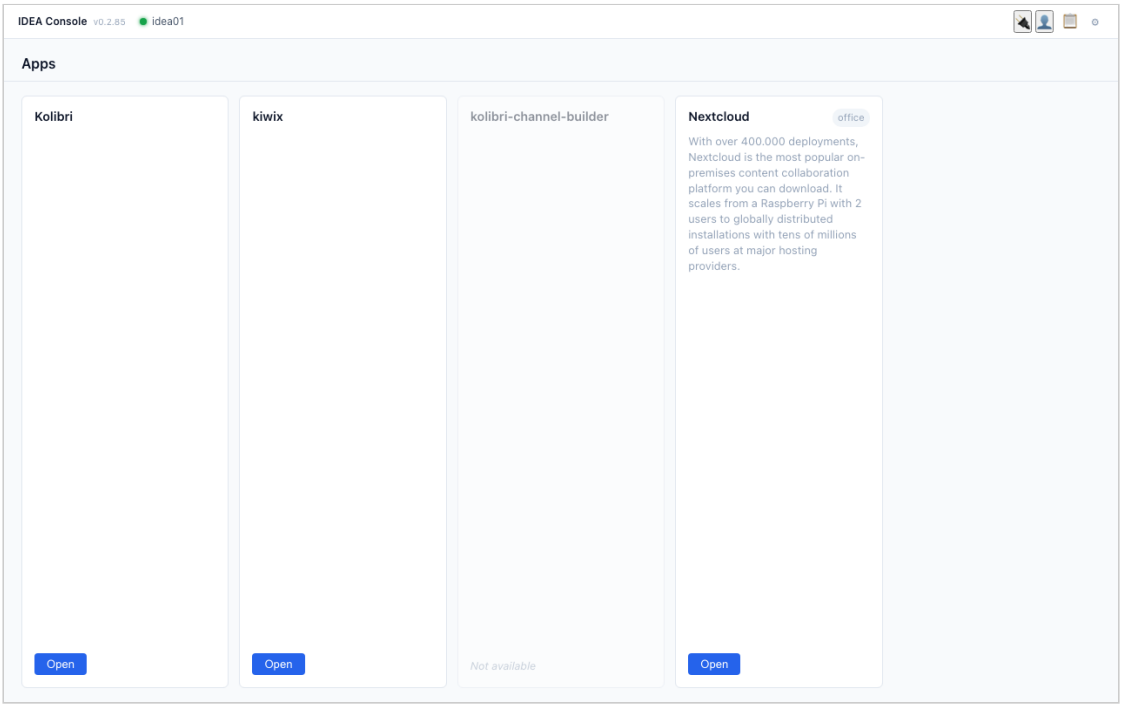


Figure 4. Console Apps page (two of the cards)

2. Click the blue **Open** button on the card of the app you need. The app opens in a new tab. Here it is Kolibri's sign-in page.

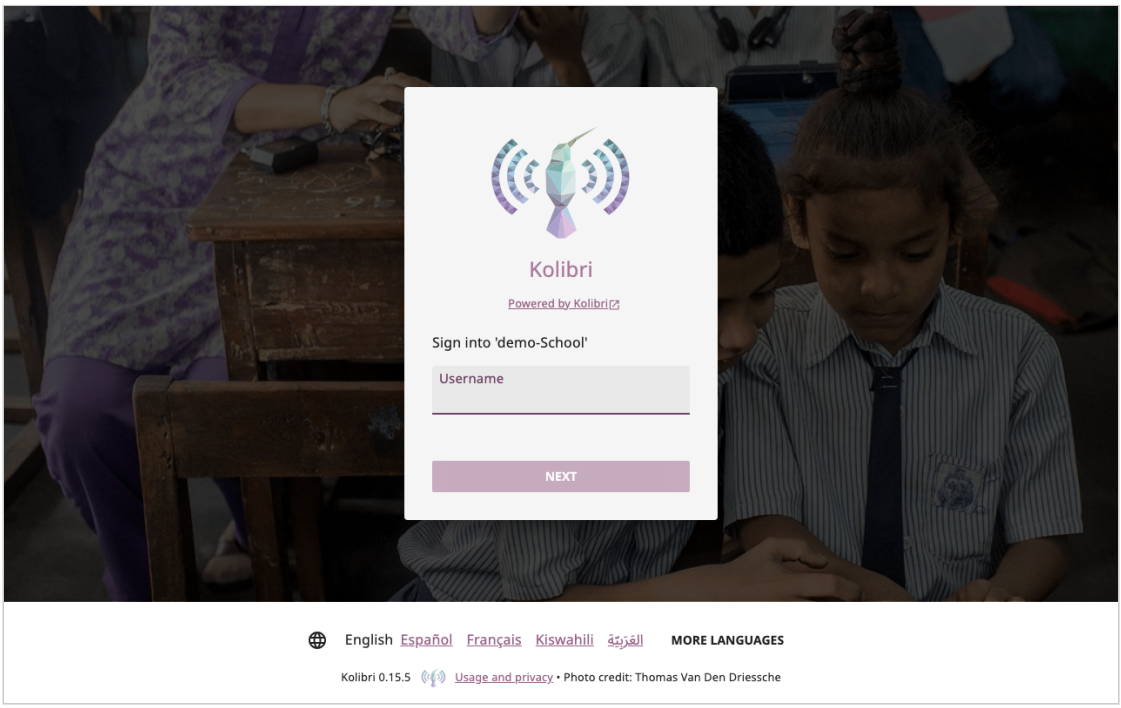


Figure 5. Kolibri opened from the Console

Good to know:

- A card that is greyed out and says **Not available** is not running. Do not try to fix it: tell your school operator.
- **Known problem: apps that don't open.** If an IDEA box is switched off, or was once connected to yours and is now at another school, the Console still shows its apps as if they worked. **Open** then gives a page that doesn't load. There is no fix yet. Use the apps that are **on** your own school's IDEA box (the name on its sticker).
- **The classroom computers reset at every restart.** Anything saved on the computer itself is gone after a restart. Save work in Nextcloud (chapter 6).
- Do not type numbers (addresses or ports) into the address bar. Always start from the Console and use **Open**.
- A student computer that no longer works properly is not repaired: the operator resets its card (IDEA for Operators: Tutorial). Students lose only what they saved on the computer itself, so let them save their work in Nextcloud (chapter 6).
- The person icon at the top right (**Operator Login**) is for the operator (IDEA for Operators: Tutorial). Teachers do not need it.

> **Tip.** You can bookmark the page that opens. If the bookmark stops working, go back to the Console and use **Open** again.

3. Kolibri for teachers

About the examples in this chapter. Every IDEA box comes with the demo setup switched on: the app **Kolibri Demo**, with the school **IDEA Demo School**, the class **Grade 5** and the demo accounts. The screens in this chapter come from it. Open the **Kolibri Demo** card in the Console, not the **Kolibri** card. The steps work the same in the school's own Kolibri, with your own school, class and account names. The demo accounts are listed in Appendix: demo accounts at the end of this tutorial.

In the demo, you sign in as `teacher1` (or `teacher2`). The demo school has three Kolibri channels: **Maths**, **Grade 5 Science** and **Early Reading**. The demo lesson uses **Maths > Fractions**. The quiz in 3.5 was made in a Kolibri with the **Khan Academy (English - US curriculum)** channel (**Math by grade > 5th grade > Add and subtract fractions**).

3.1 Signing in

Open **Kolibri Demo** from the Console (2.3). Type your username and click **NEXT**, then type your password.

3.2 Create a class

Creating classes needs a Kolibri admin account (in the demo school, `admin`, password `admin911!`; `teacher1` is the coach of Grade 5).

1. Open **Facility > CLASSES**. You see the list of classes.

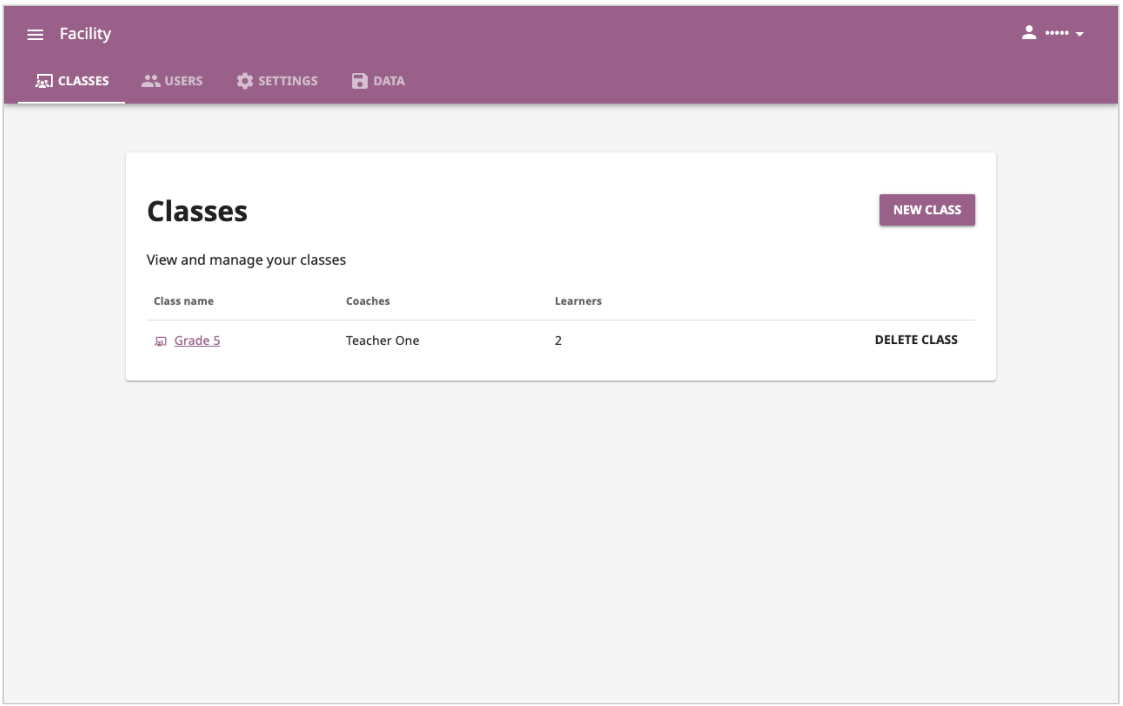


Figure 6. Classes list

2. Click **NEW CLASS**, type the class name, click **SAVE**. (In the demo we pressed **CANCEL** because `Grade 5` already exists.)

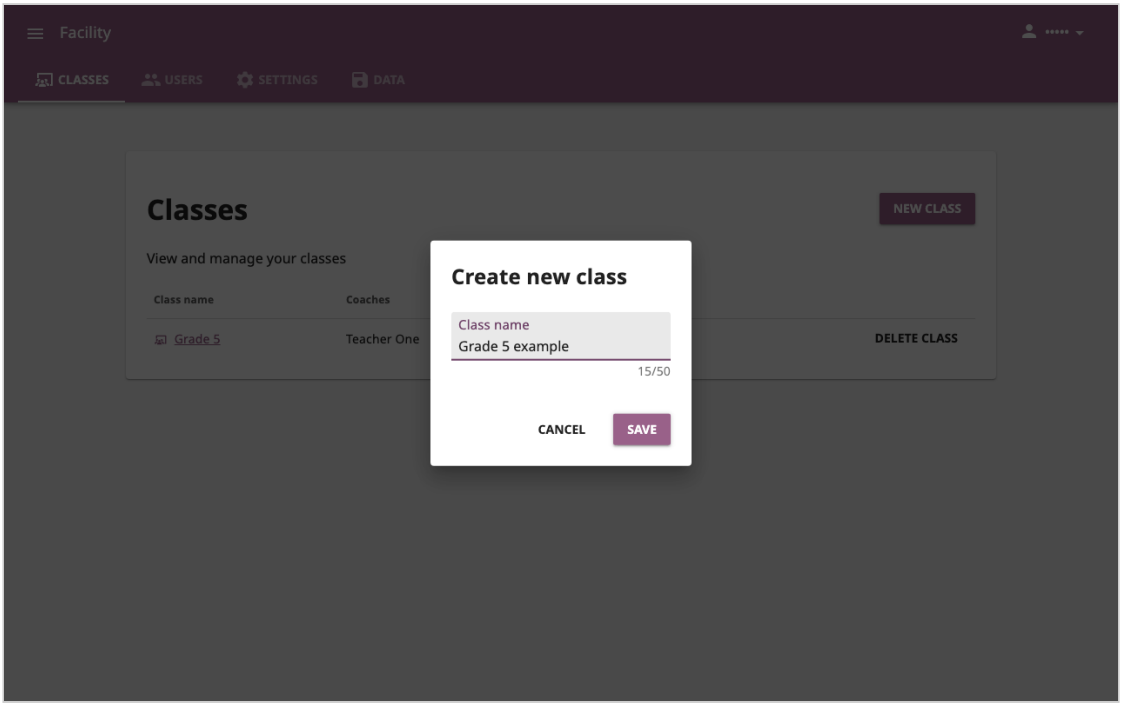


Figure 7. Create new class

3.3 Enroll learners

1. Click the class name. On the class page (with **Coaches** and **Learners**), click **ENROLL LEARNERS**. The **Enroll learners** page opens.

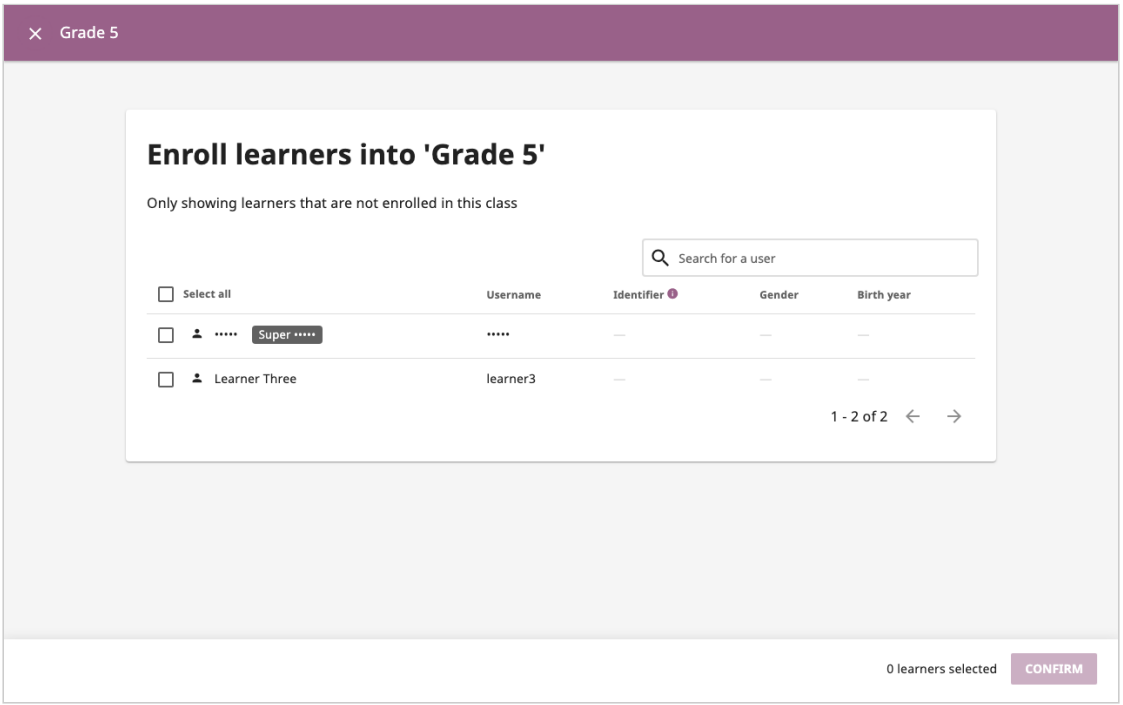


Figure 8. Enroll learners page

2. The list only shows learners who are not yet in the class. Tick each learner (here: Learner Three), then click **CONFIRM**.

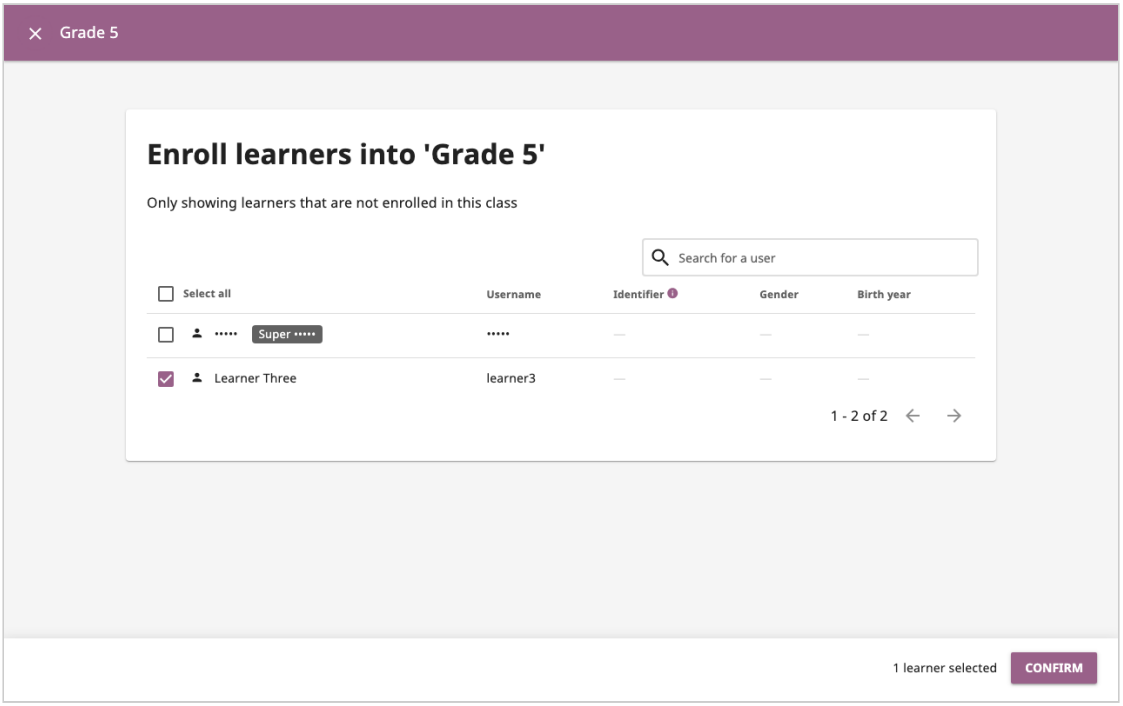


Figure 9. Learner ticked

3. The learner now appears under Learners.

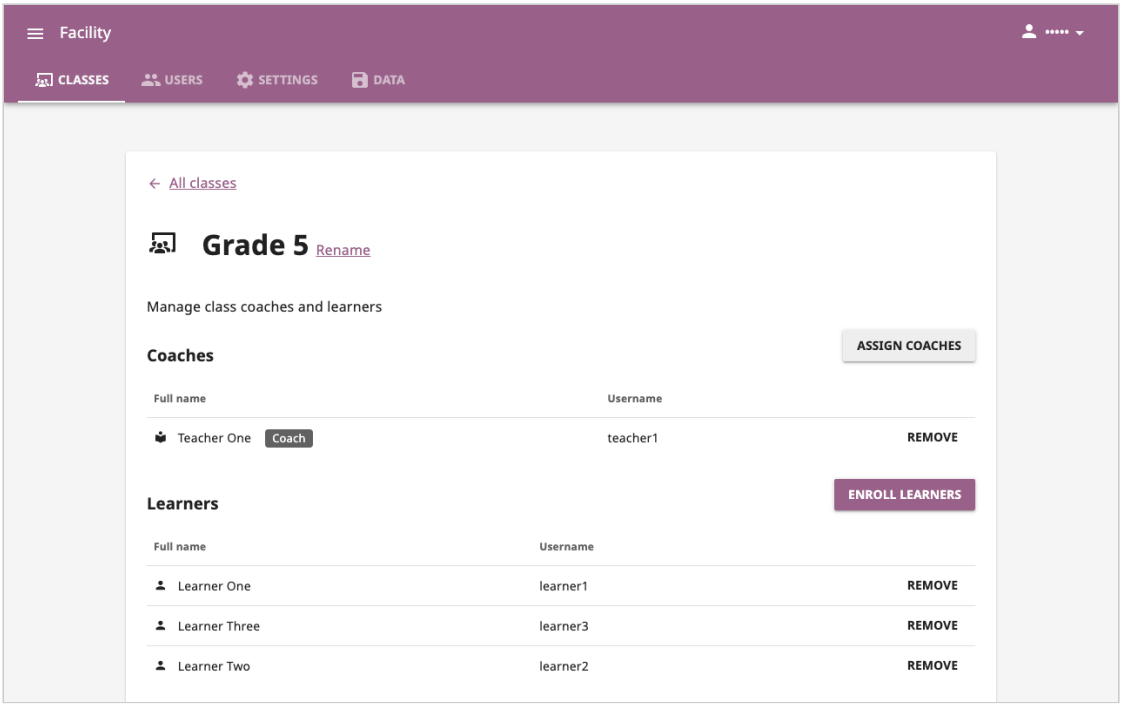


Figure 10. Learner enrolled

3.4 Make a lesson and show it to the class

A good lesson mixes resources that explain (videos, activities) with resources that let learners practise (exercises).

1. Open **Coach**, choose the class, click **PLAN > LESSONS > NEW LESSON**.

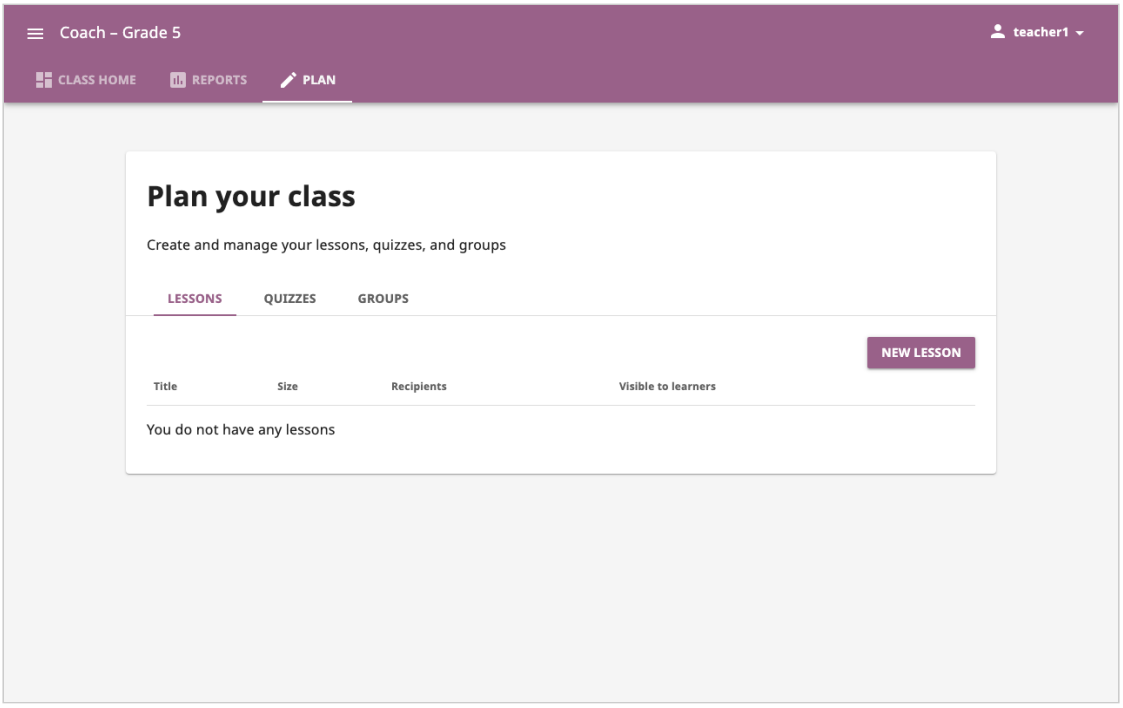


Figure 11. Plan your class

2. Type a title (`Fractions lesson`). Under **Recipients** keep **Entire class**. Click **SAVE CHANGES**.

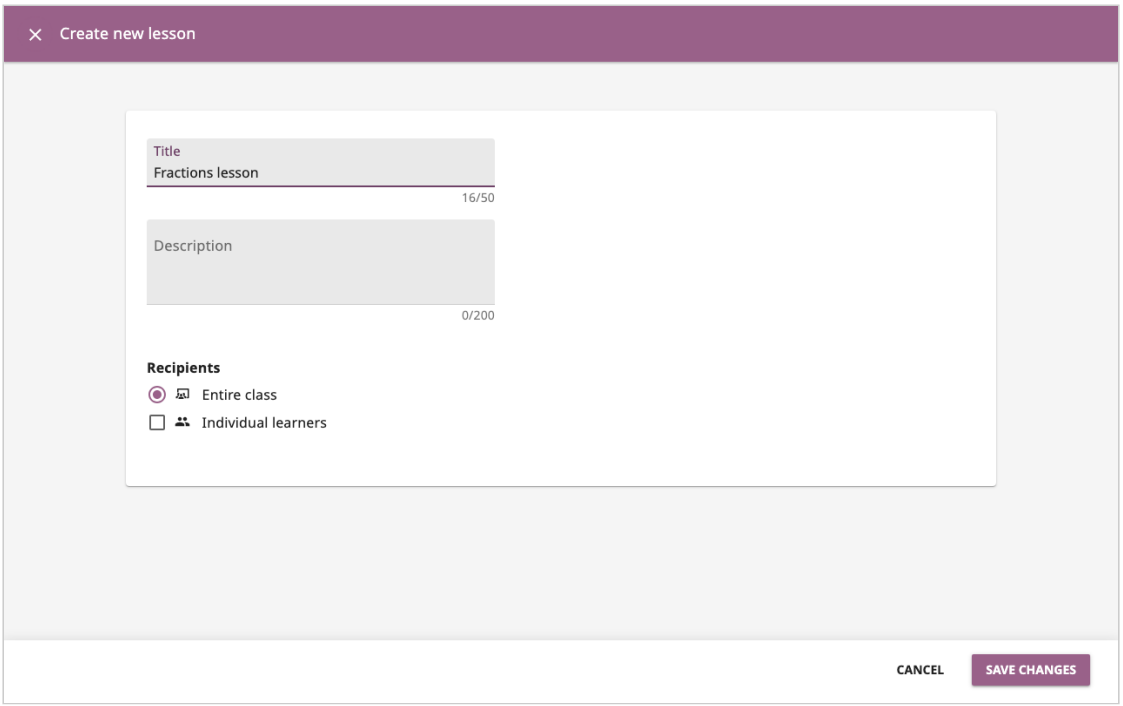


Figure 12. New lesson

3. On the lesson page, click **MANAGE RESOURCES**.

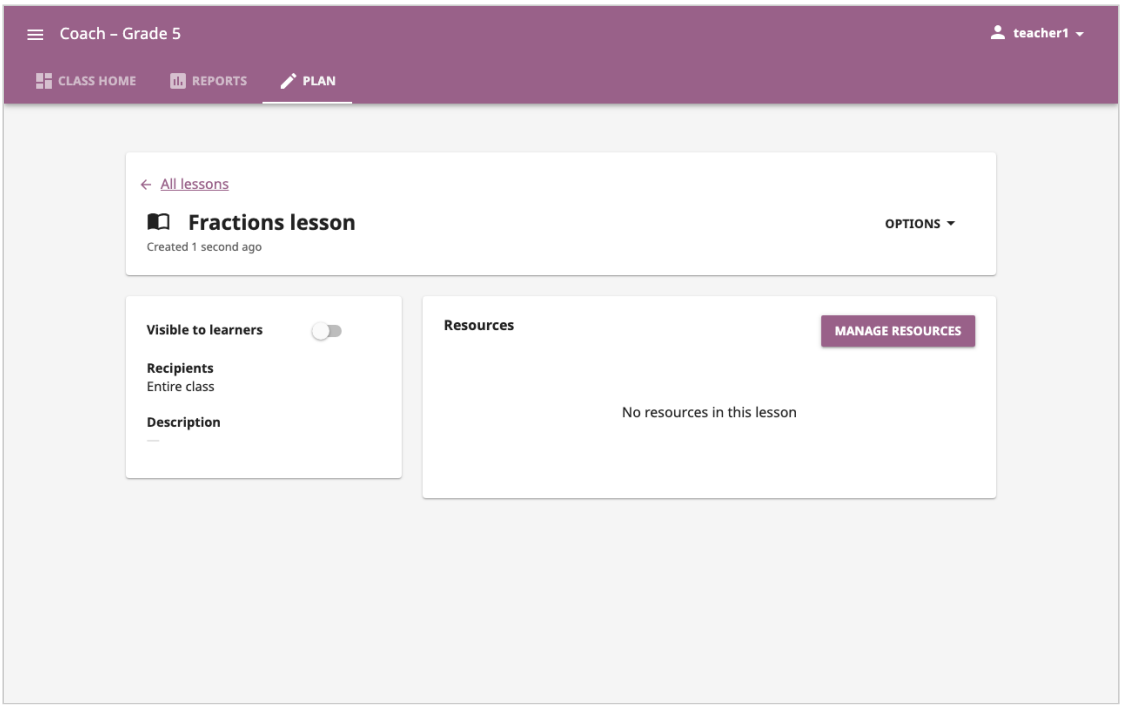


Figure 13. Empty lesson

4. Choose the channel.

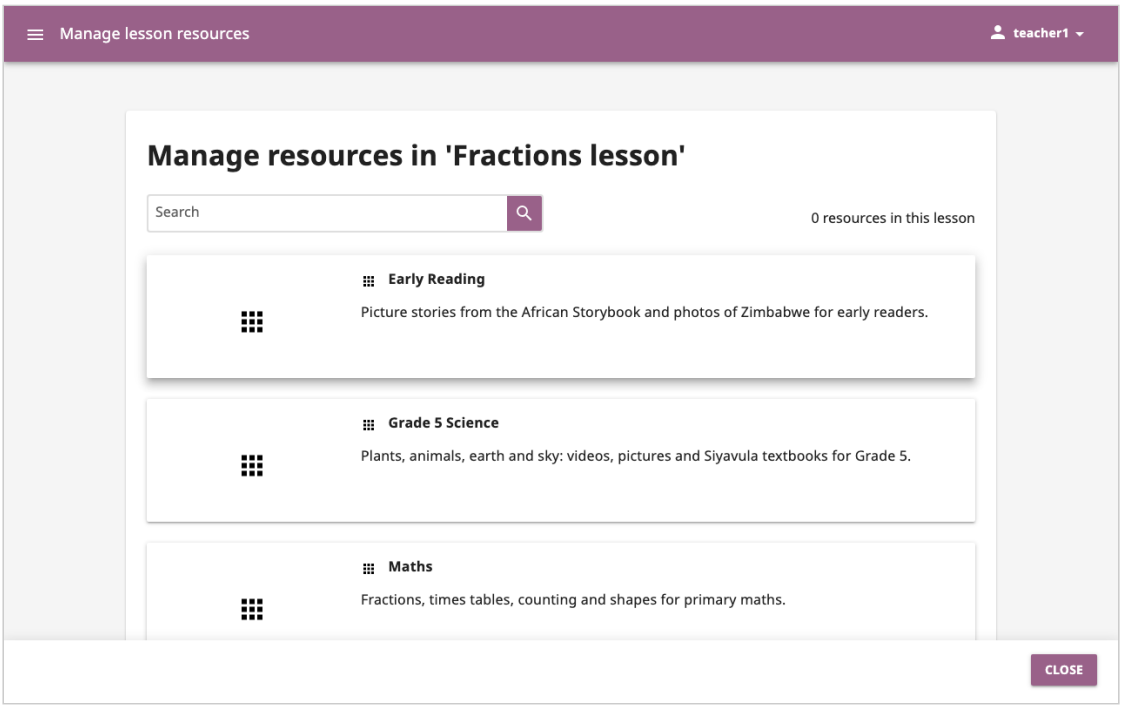


Figure 14. Channels

5. Click the channel (**Maths**). You see its topics.

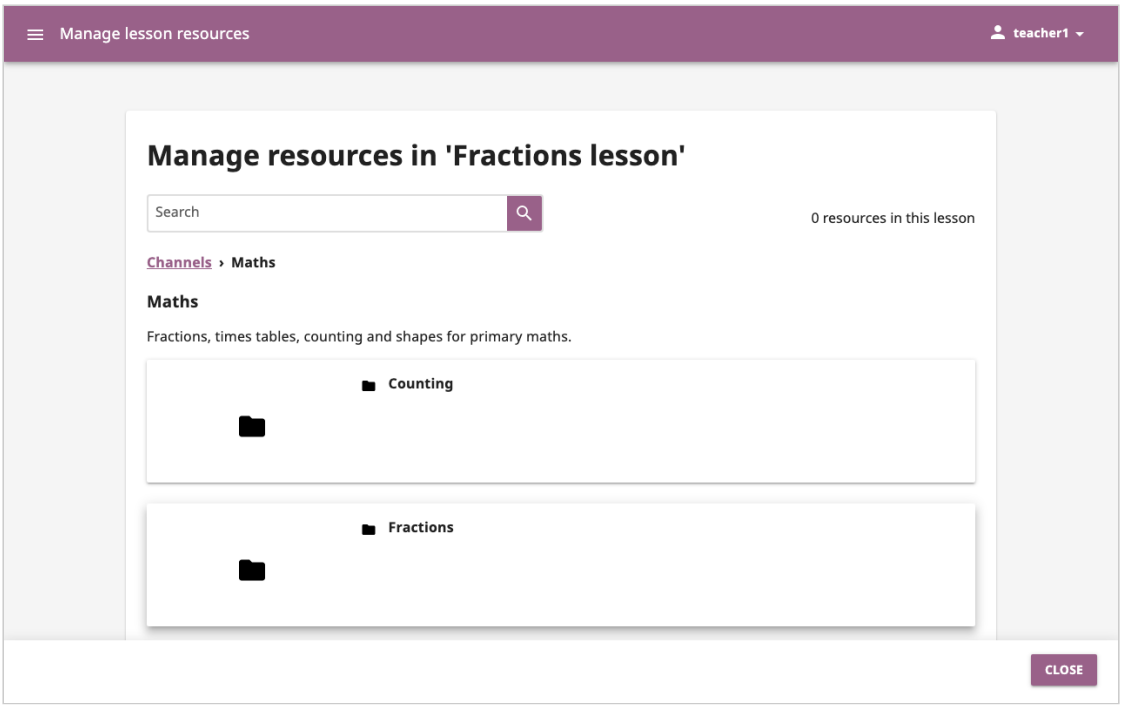


Figure 15. Topic

6. Open the topic **Fractions** and tick the resources you want (here all 3: Cake in quarters, Fraction circles, Paper fraction pies). Each tick is saved straight away; the count at the top right goes up. Click **SAVE CHANGES** (or **CLOSE**) when done.

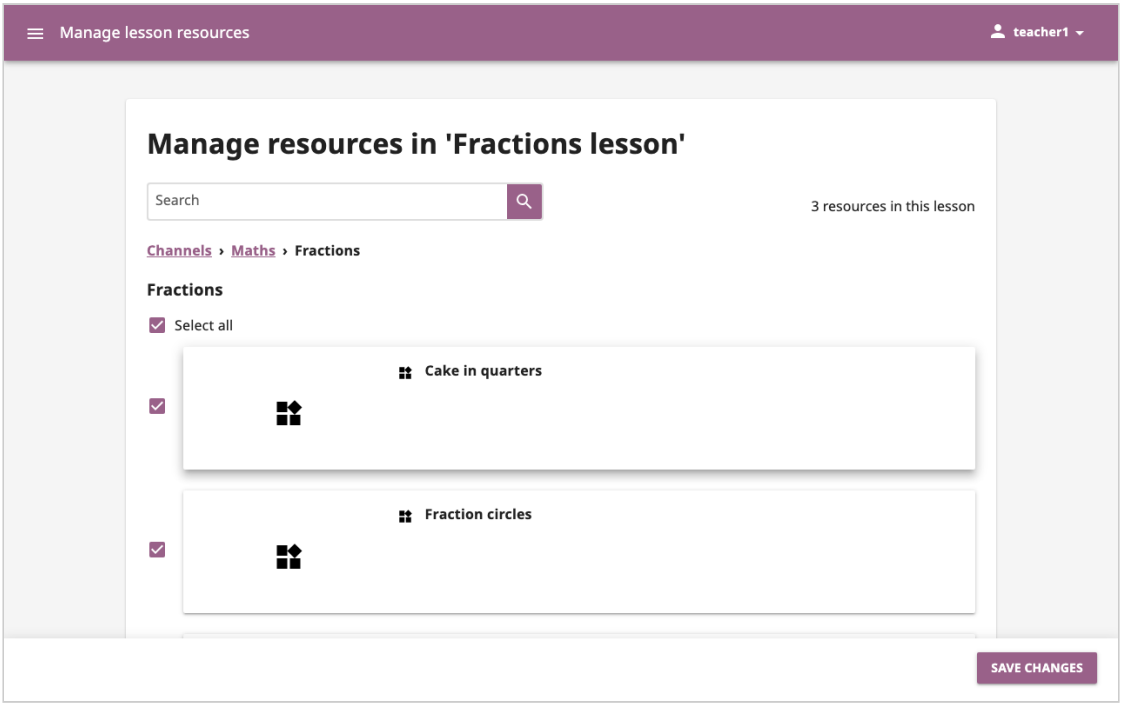


Figure 16. Resources ticked

7. Back on the lesson page, check the resources and switch on **Visible to learners**.

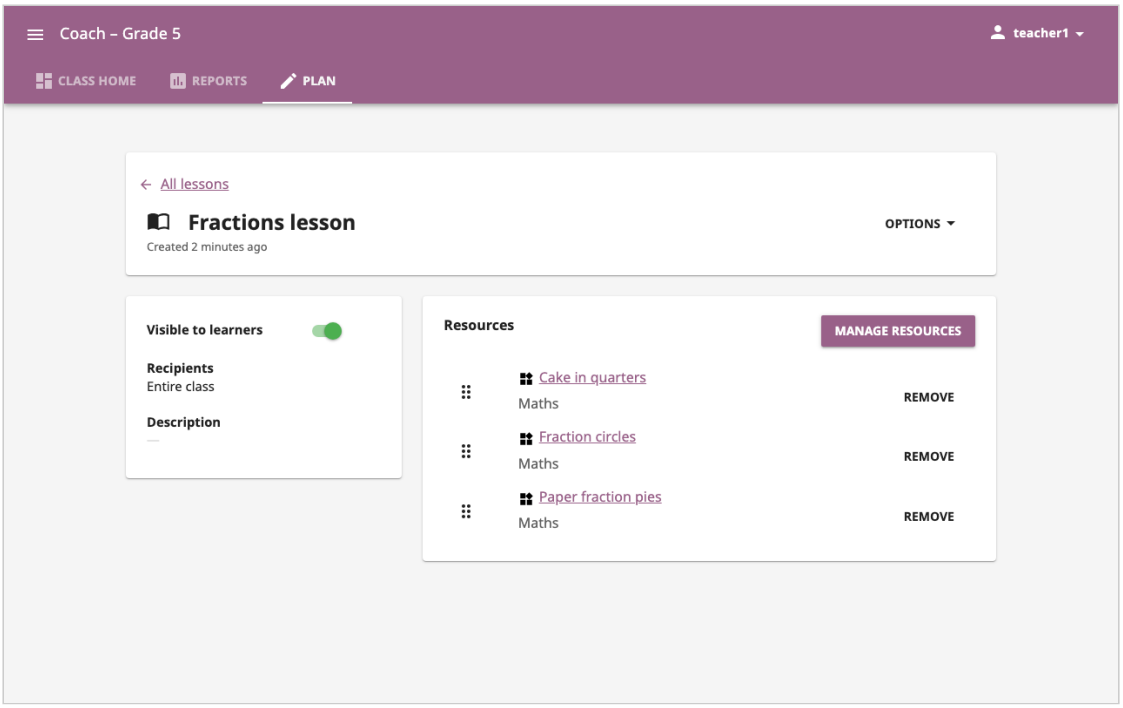


Figure 17. Lesson with its resources, visible

3.5 Make a quiz

Quizzes are built from exercises only.

Try it in Kolibri Demo

Kolibri Demo has exercises and a quiz ready for the class **Grade 5**: in **Maths**, **Fractions practice** and **Times tables practice** (6 questions each; the goal is 4 correct out of the last 5); in **Early Reading**, **Reading: Rudo and the goat** (4 questions, all must be right); and the quiz **Grade 5 Practice Quiz** (10 questions). Try them first as a learner, then see the results as the teacher. Use a second browser window (private window) for the learner.

A. A learner does an exercise

1. **Do:** sign in to Kolibri Demo as `learner1` (no password). Under **Recent lessons**, click **Maths lesson**.
   ✔ **Check: HOME** shows the class **Grade 5**, three lessons and the quiz **Grade 5 Practice Quiz**.

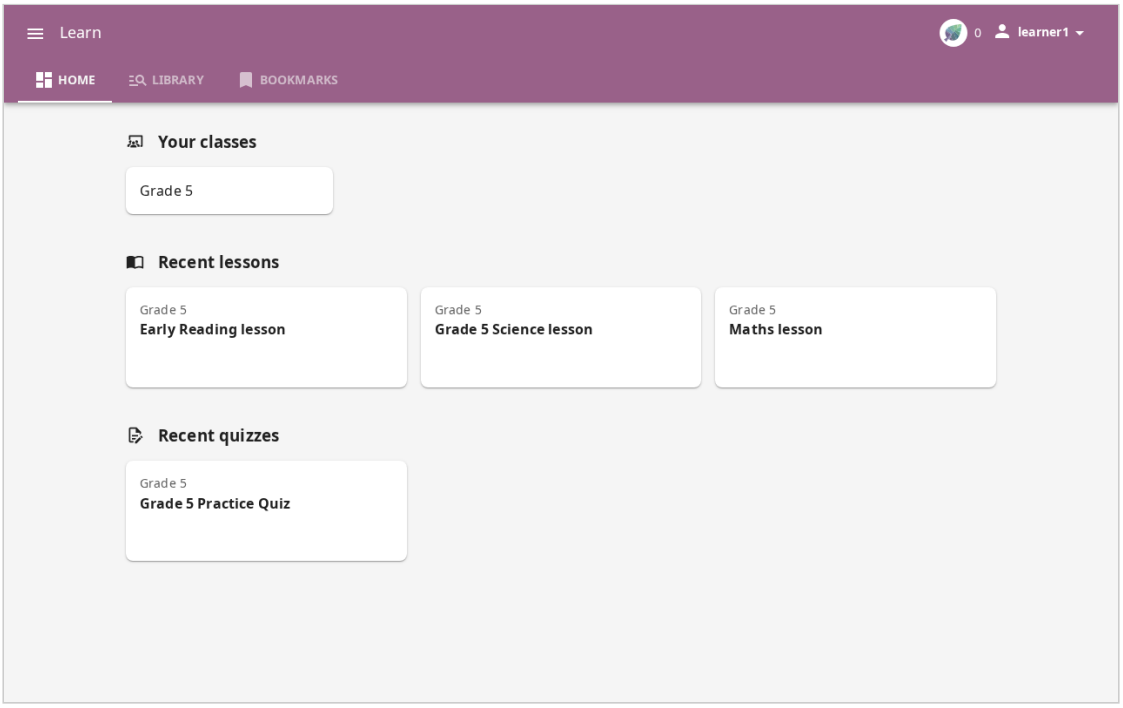


Figure 18. Learner home in Kolibri Demo

2. **Do:** in the lesson, click **Fractions practice** (marked **Practice**).
   ✔ **Check:** the lesson lists its items in order: videos, pictures, documents and the two practice exercises.

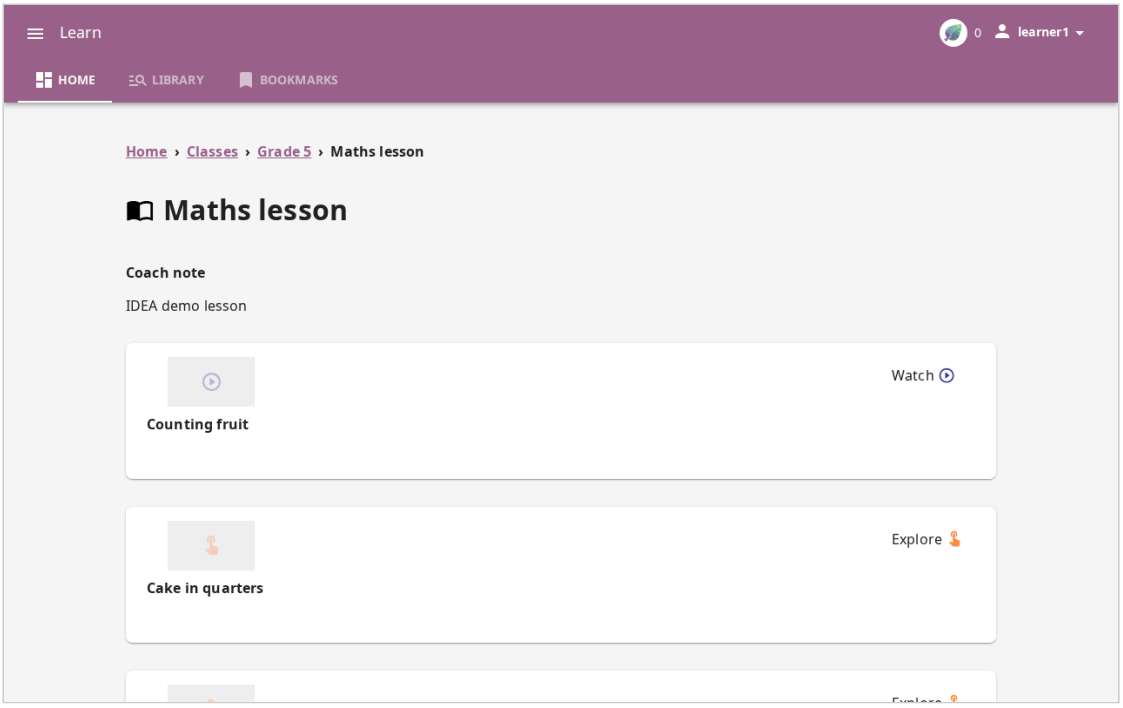


Figure 19. Maths lesson

3. **Do:** read the question, click an answer (or type a number), then click **CHECK**.
   ✔ **Check:** the goal **Get 4 correct** is shown at the top and bottom.

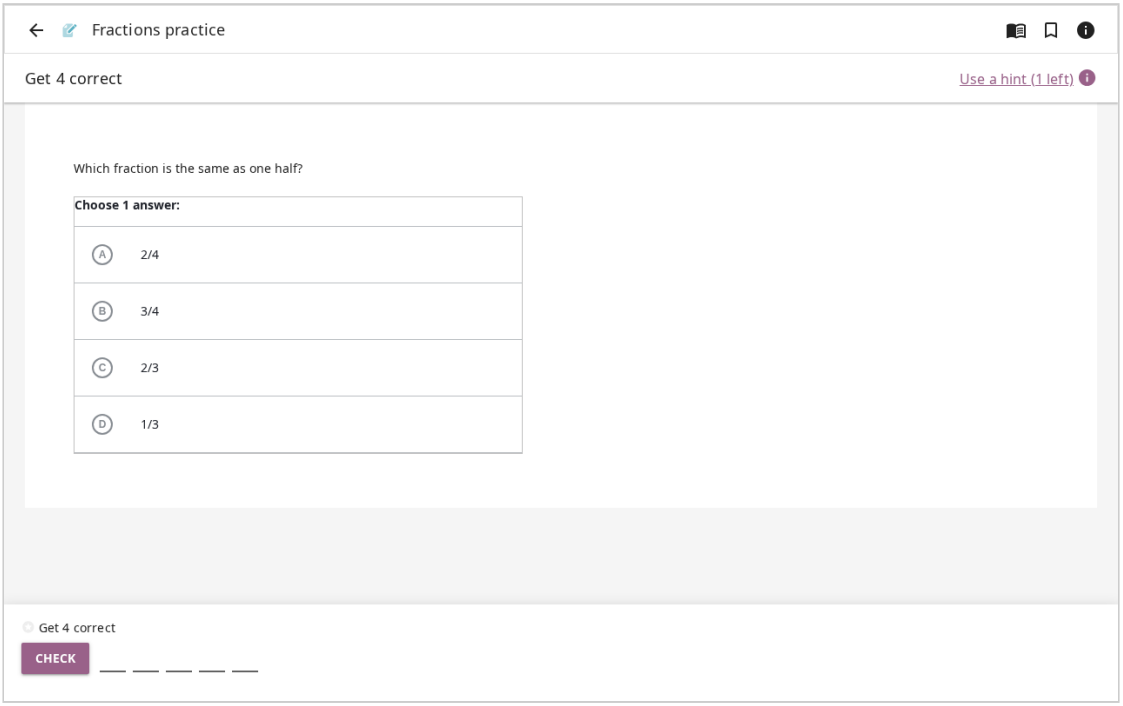


Figure 20. Exercise question

4. **Do:** after a right answer, click **NEXT**. After a wrong answer, try again, then go on.
   ✔ **Check:** a green tick appears at the bottom for each right answer.

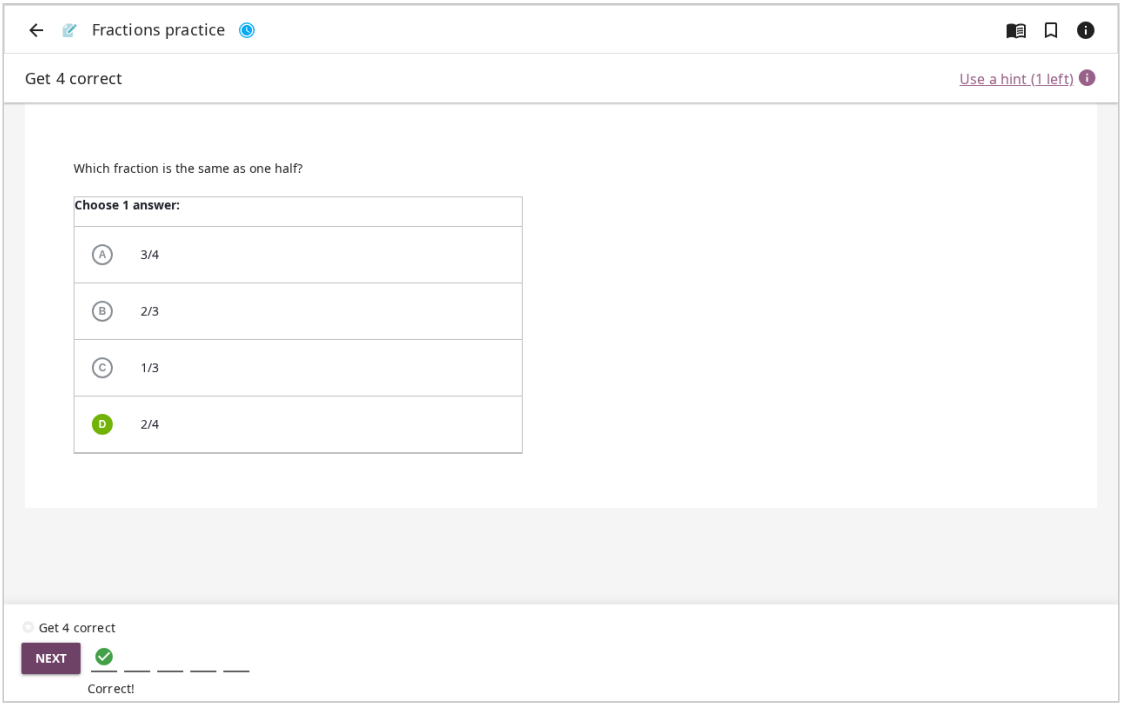


Figure 21. Right answer

5. **Do:** keep going until the goal is reached.
   ✔ **Check: Resource completed** opens, with points. Click **MOVE ON** for the next item, or close the window.

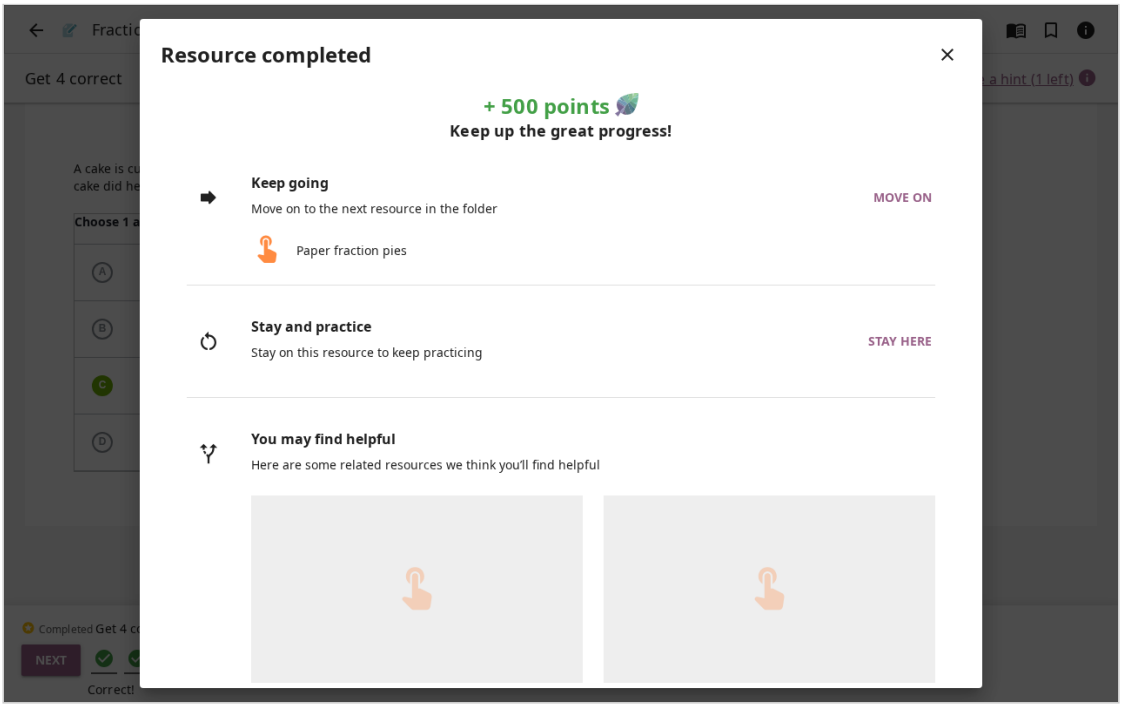


Figure 22. Exercise completed

B. A learner takes the quiz

1. **Do:** still as `learner1`, go to **HOME** and click **Grade 5 Practice Quiz** under **Recent quizzes**.
   ✔ **Check:** the quiz opens at **Question 1 of 10**. The list on the left shows all questions.

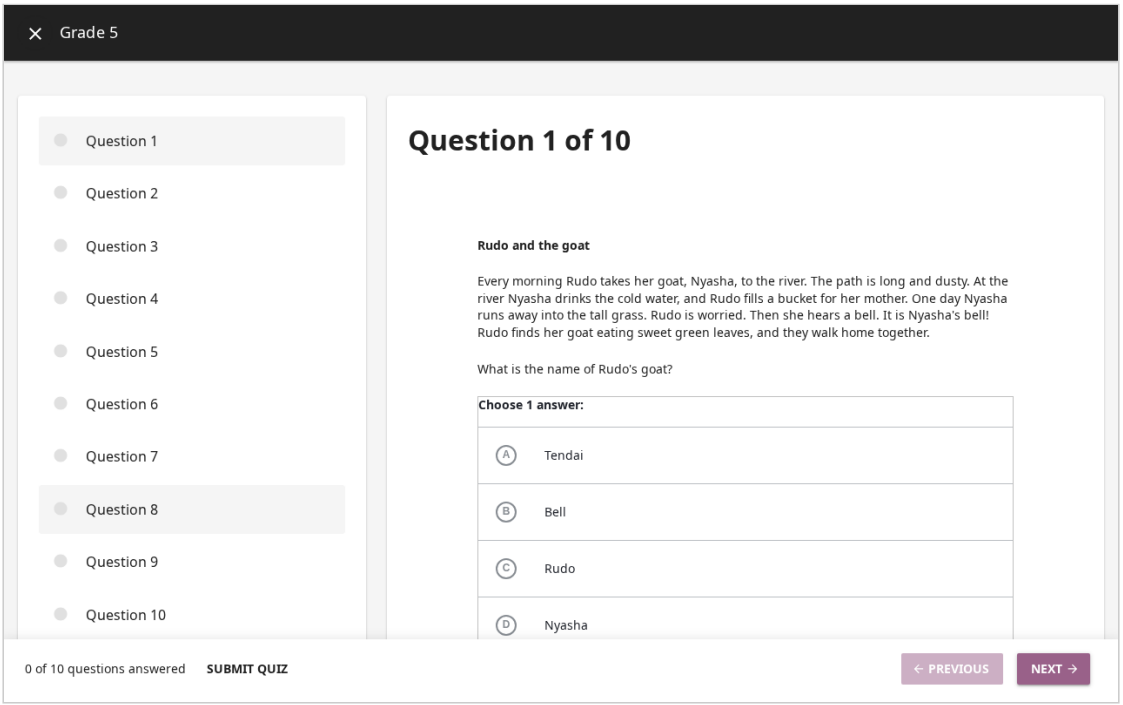


Figure 23. Quiz question

2. **Do:** answer each question and use **NEXT** / **PREVIOUS**. Some questions have more than one right answer (**Choose all answers that apply**).
   ✔ **Check:** answered questions get a blue dot in the list, and the bottom shows **10 of 10 questions answered**.

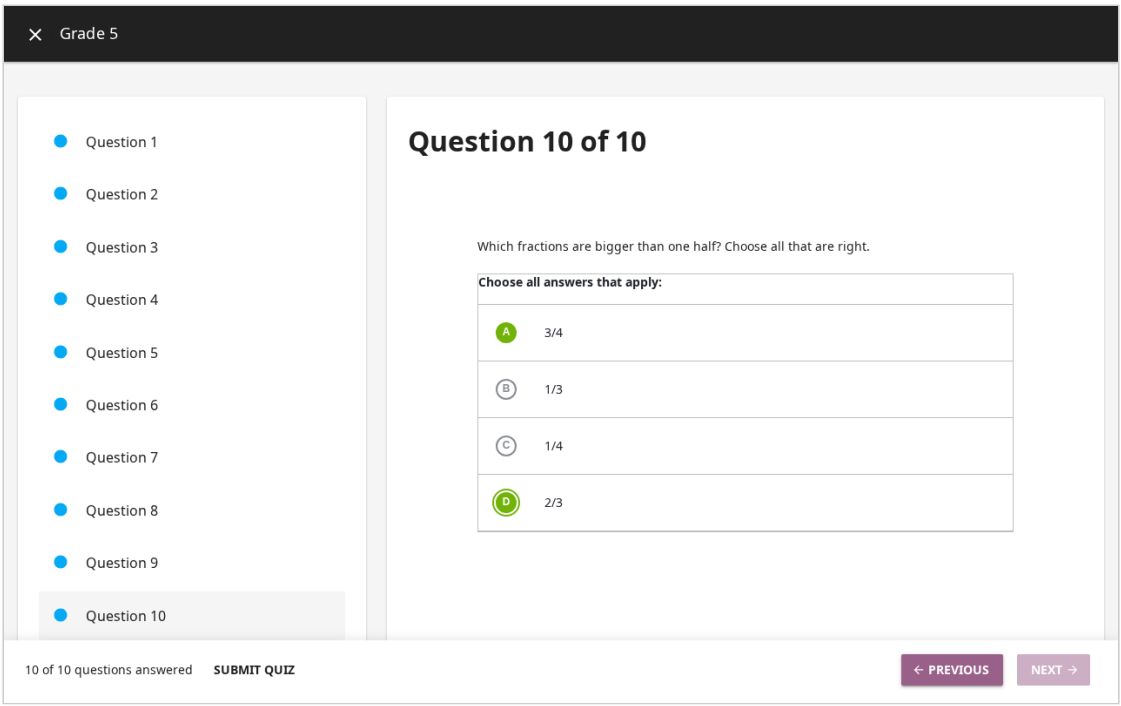


Figure 24. Last question

3. **Do:** click **SUBMIT QUIZ**, then **SUBMIT QUIZ** again in the window that opens. Answers cannot be changed after that.

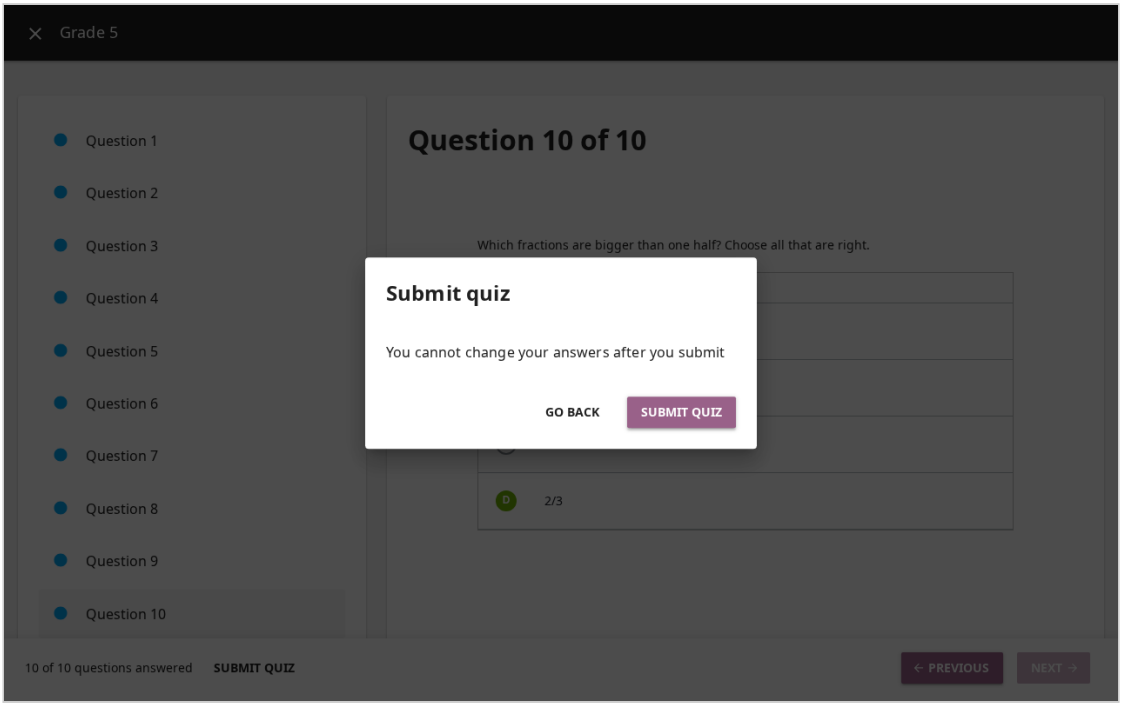


Figure 25. Submit quiz

✔ **Check:** the class page shows the quiz with its score, and the lesson as **In progress**.

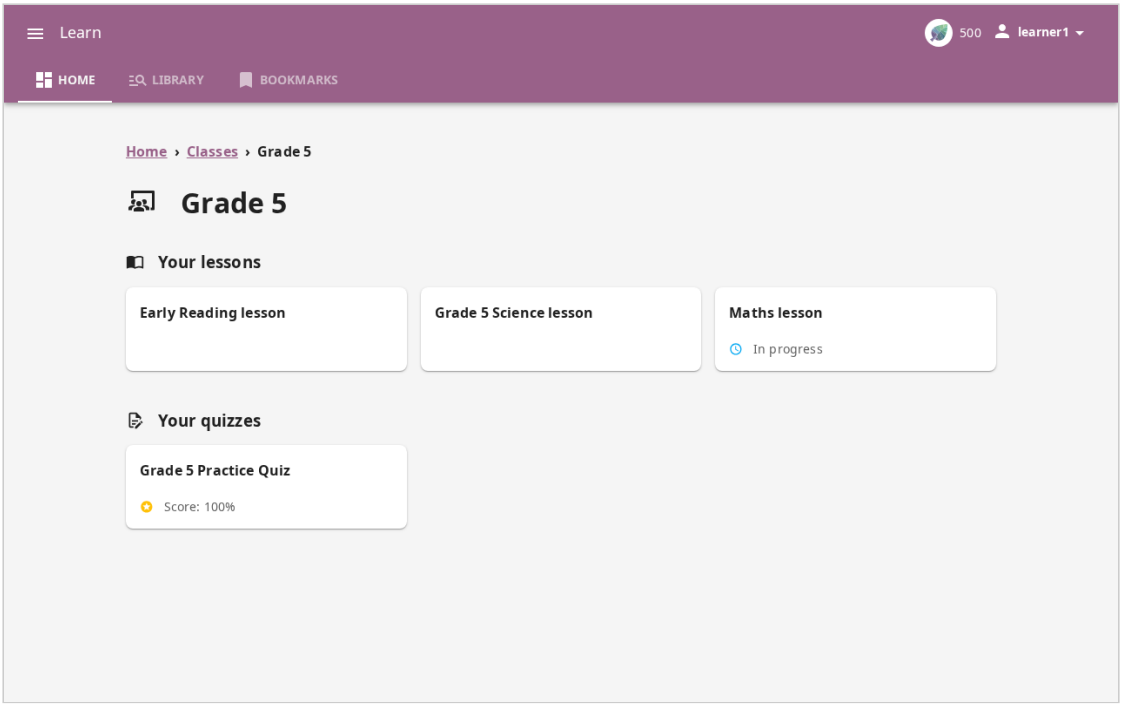


Figure 26. Quiz score

C. The teacher sees progress

1. **Do:** in the teacher window, sign in to Kolibri Demo as `teacher1` and open **Coach > Grade 5**.
   ✔ **Check: CLASS HOME** shows how many learners completed the quiz and started each lesson. **Class activity** lists what learners did.

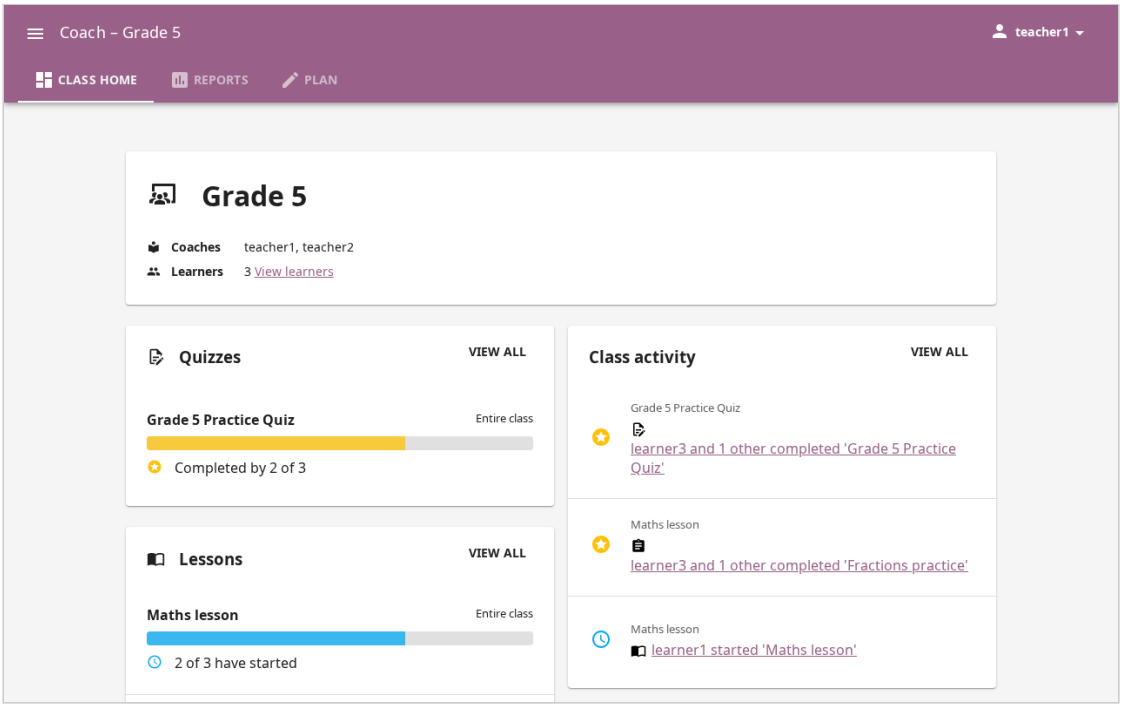


Figure 27. Coach class home

2. **Do:** click **REPORTS > LESSONS > Maths lesson**, then **Fractions practice**.
   ✔ **Check:** each learner is listed as **Completed**, **Started** or **Has not started**, with the time spent.

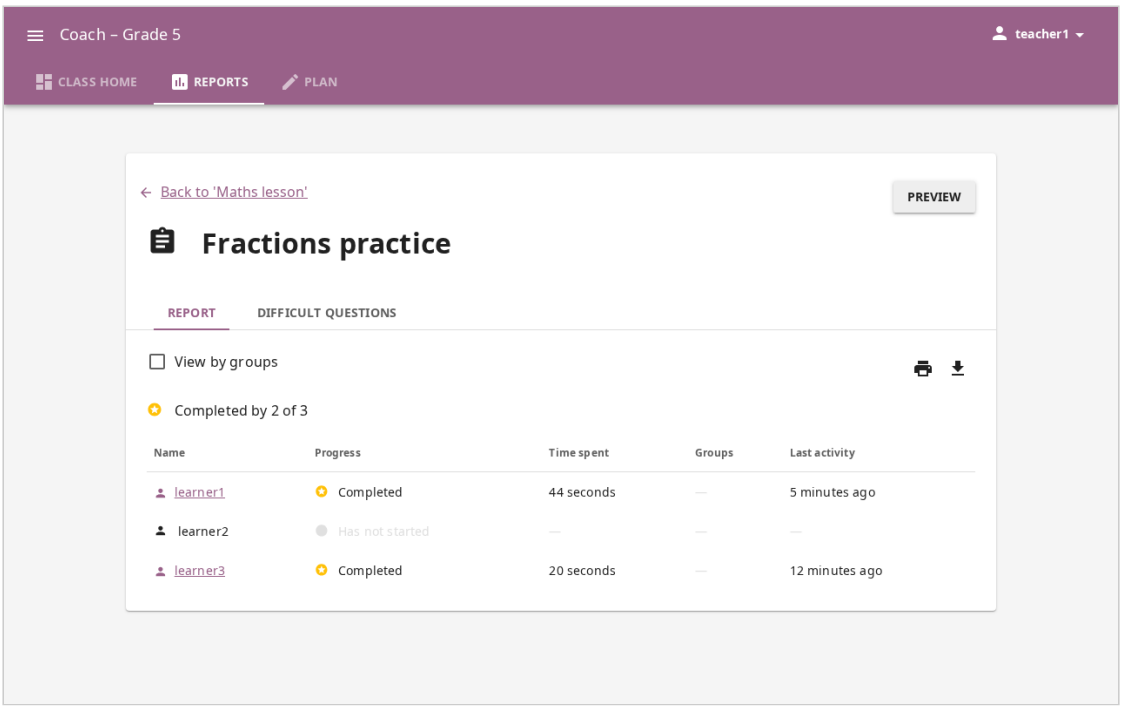


Figure 28. Exercise report

3. **Do:** click **REPORTS > QUIZZES > Grade 5 Practice Quiz**.
   ✔ **Check:** each learner's progress and score. **DIFFICULT QUESTIONS** shows the questions many learners got wrong.

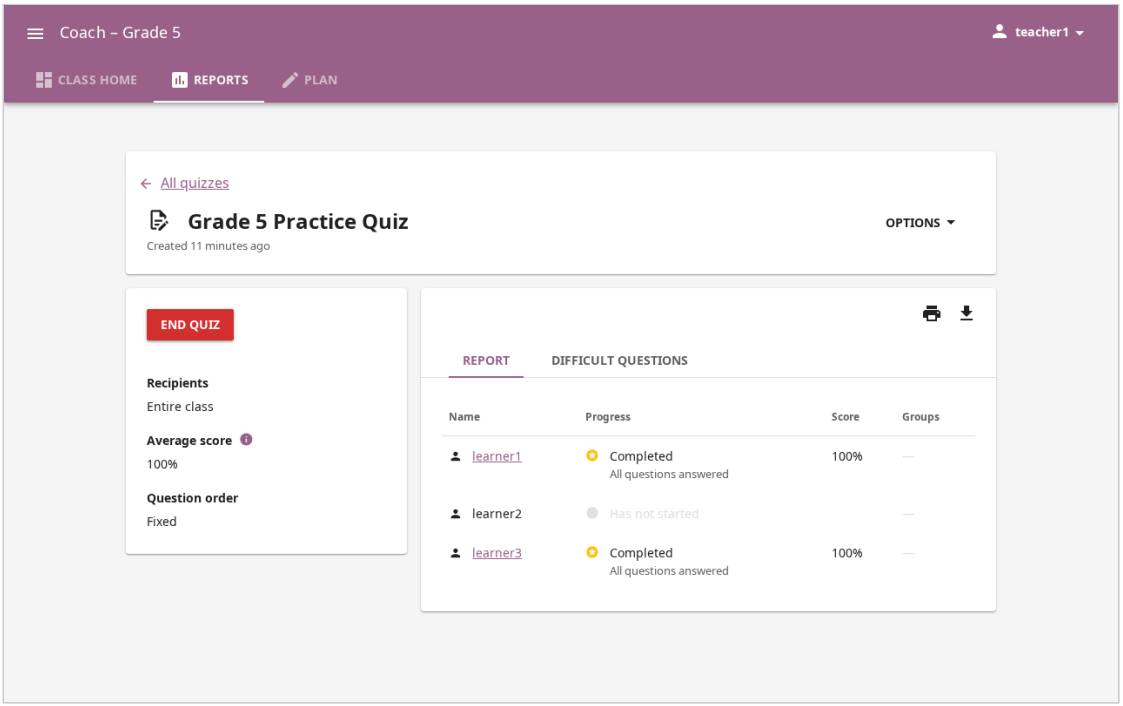


Figure 29. Quiz report

In Kolibri Demo, `learner3` has already done the exercise and the quiz, so the reports are never empty. 3.6 explains the reports in more detail.

Make your own quiz

The steps are the same in Kolibri Demo and in the school's own Kolibri. The screenshots below come from a Kolibri with the Khan Academy channel.

1. Click **PLAN > QUIZZES > NEW QUIZ**.

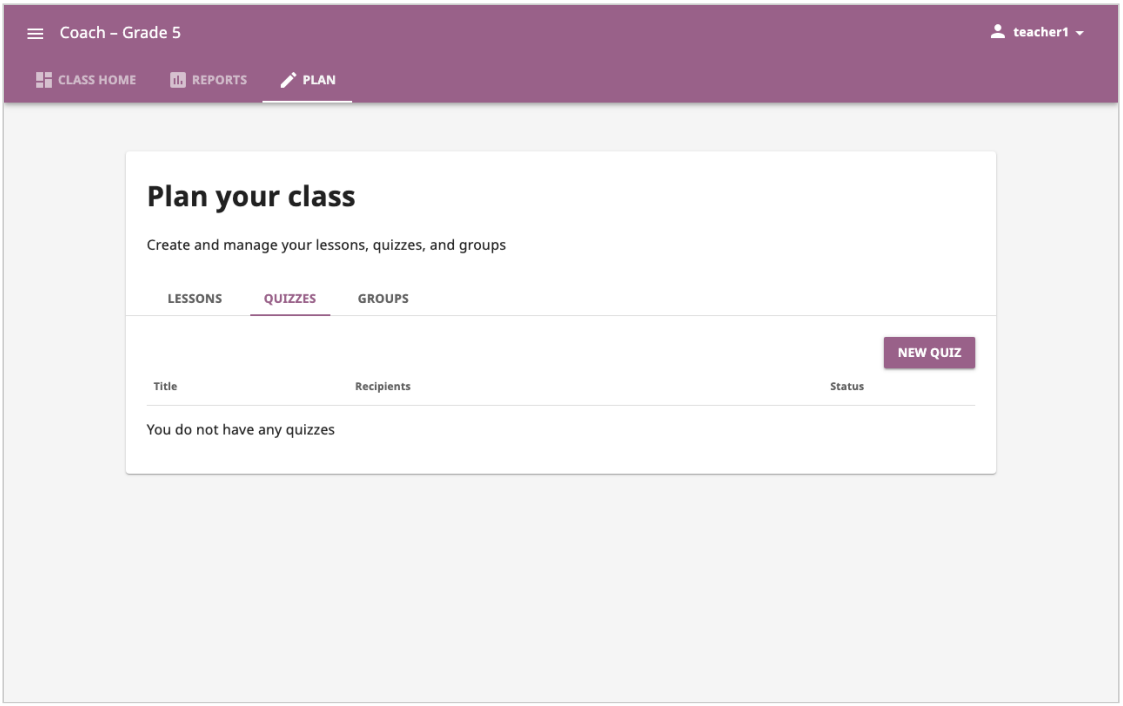


Figure 30. Quizzes tab

2. Type a title (`Fractions quiz`) and the **Number of questions** (6). Browse to **Khan Academy (English - US curriculum) > Math by grade > 5th grade > Add and subtract fractions** and tick a folder (here **Adding and subtracting fractions with unlike denominators**, 3 exercises) or single exercises. Click **CONTINUE**.

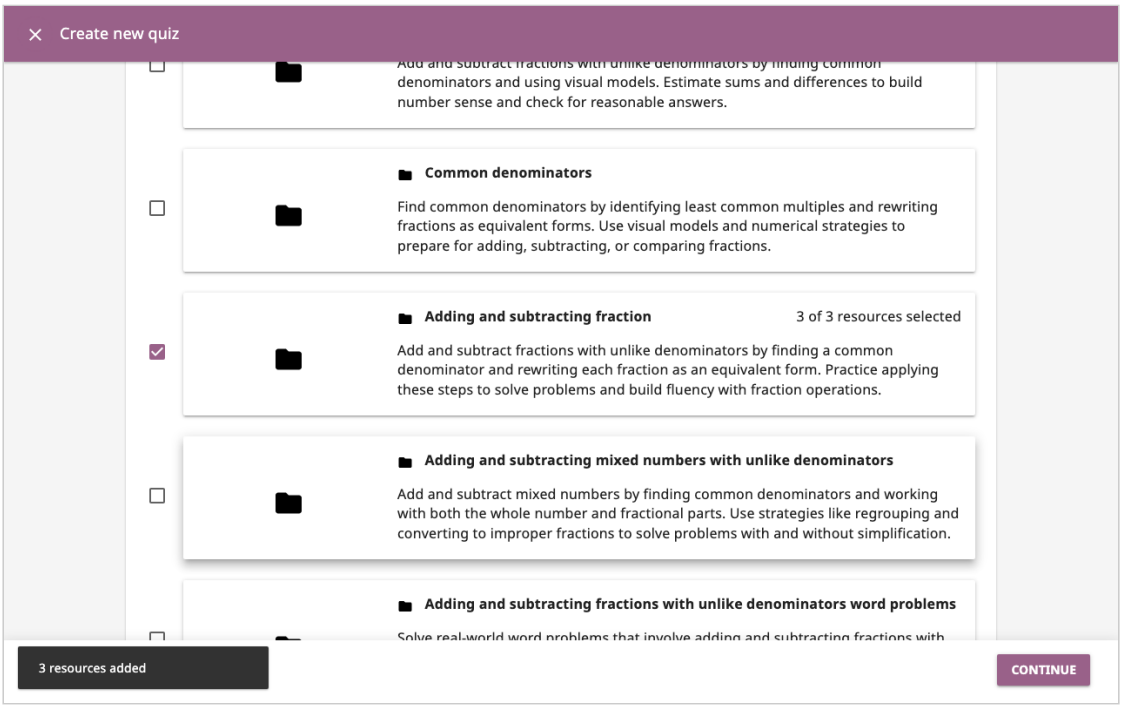


Figure 31. Create new quiz

3. **Preview quiz**: check the questions, choose **Randomized** or **Fixed** order, click **FINISH**.

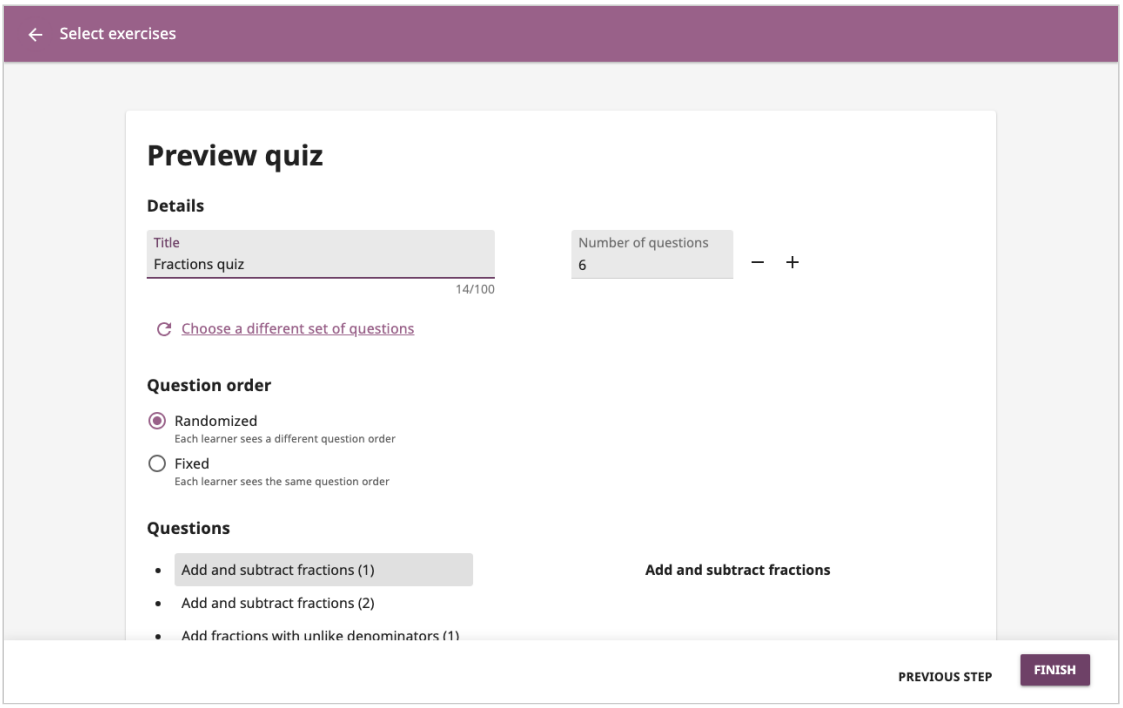


Figure 32. Preview quiz

4. The quiz is not open yet. When the class is ready, click **START QUIZ** and then **CONTINUE**. When the class has finished, open the quiz in **Coach** again and end it there, so no more answers come in.

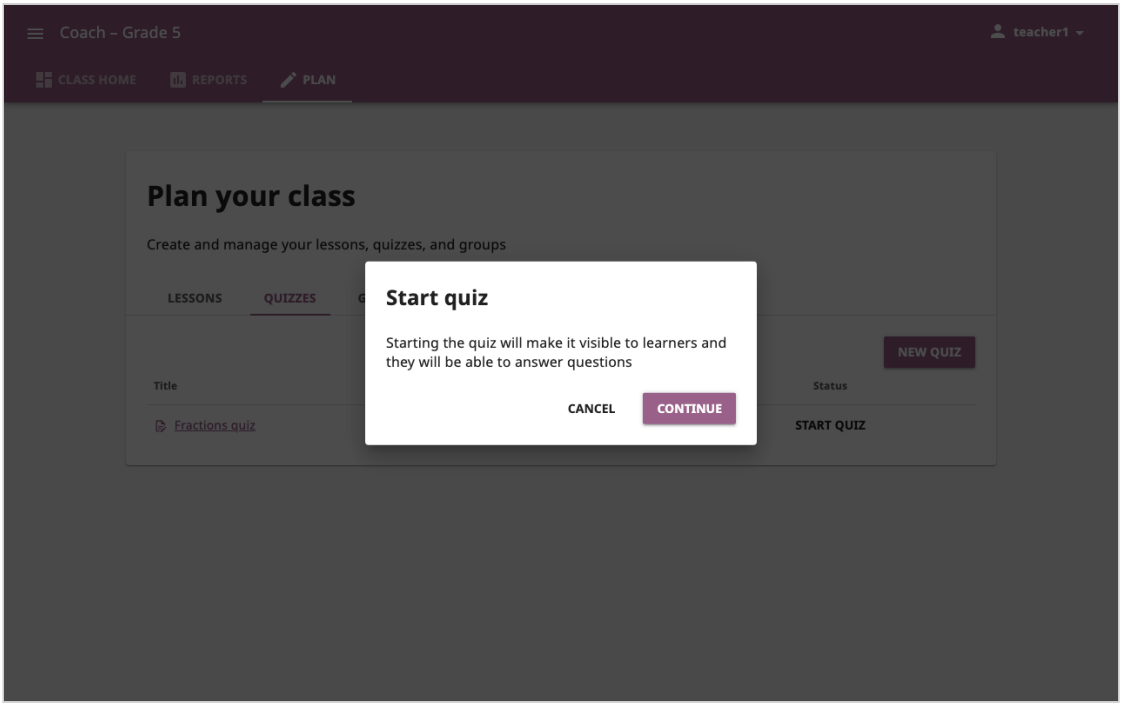


Figure 33. Start quiz

3.6 See progress (Reports)

Do this after learners have worked (chapter 4).

1. Click **REPORTS**. Tabs: **LESSONS**, **QUIZZES**, **GROUPS**, **LEARNERS**.

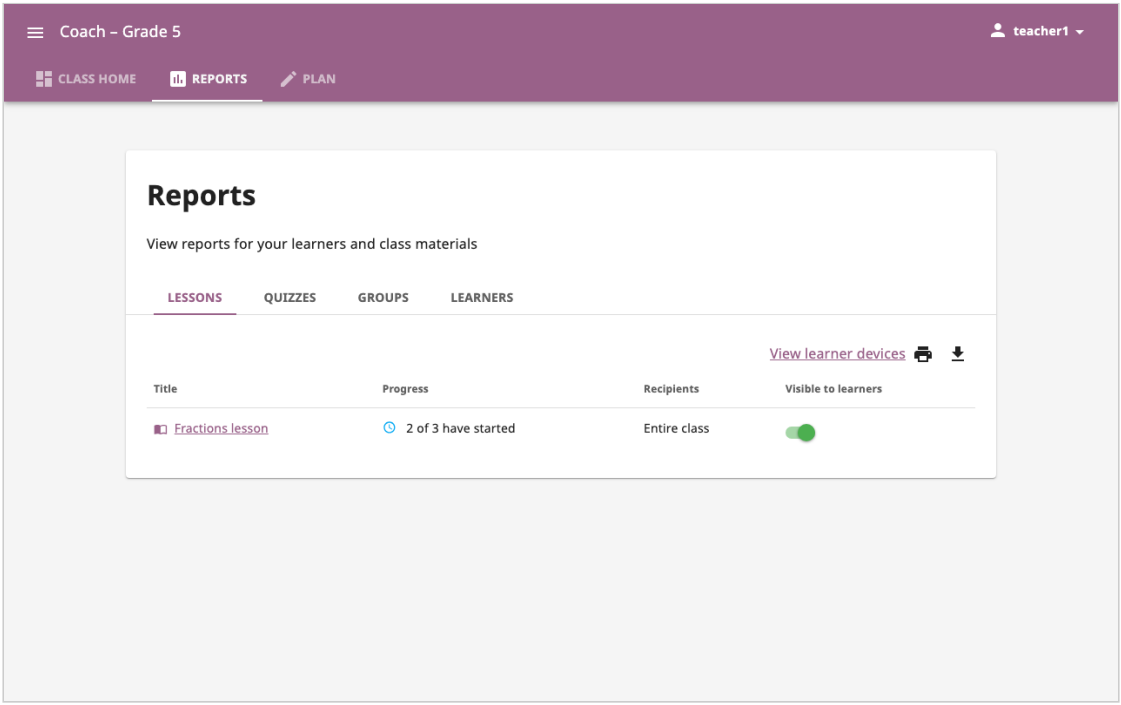


Figure 34. Reports

2. **LESSONS > (the lesson) > LEARNERS**: who completed, started or did not start.

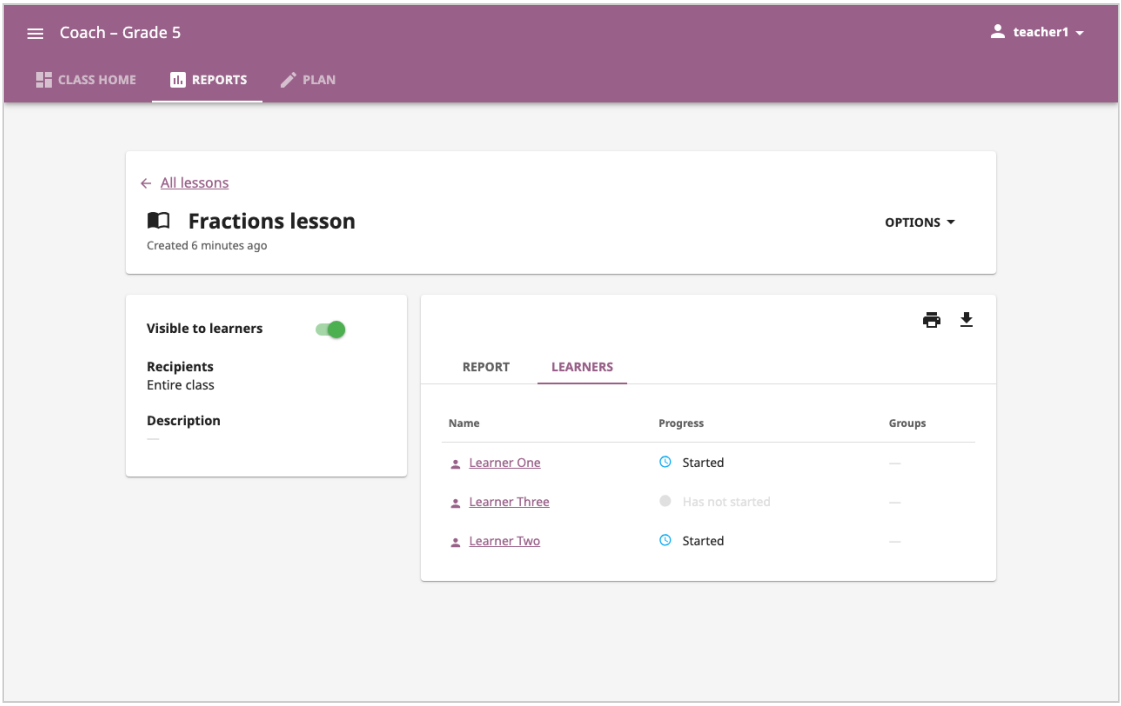


Figure 35. Lesson report

3. **QUIZZES > (the quiz)**: status and score per learner; **DIFFICULT QUESTIONS** shows what to re-teach.

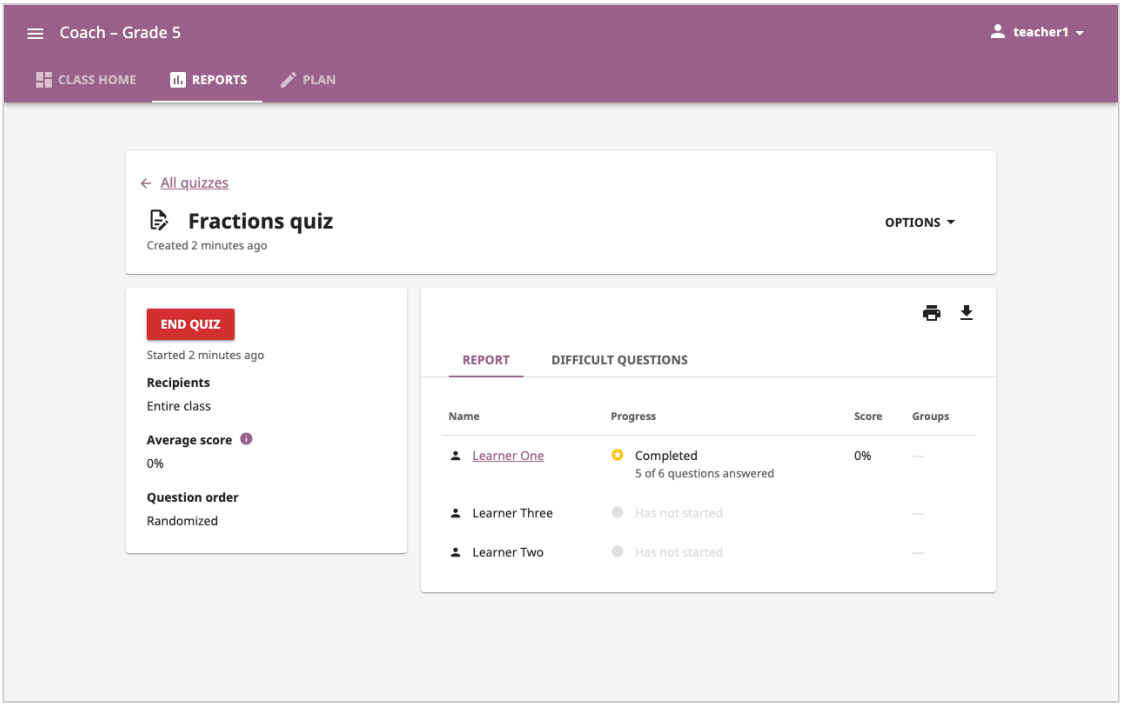


Figure 36. Quiz report

4. **LEARNERS > (a name)**: that learner's lessons and quizzes.

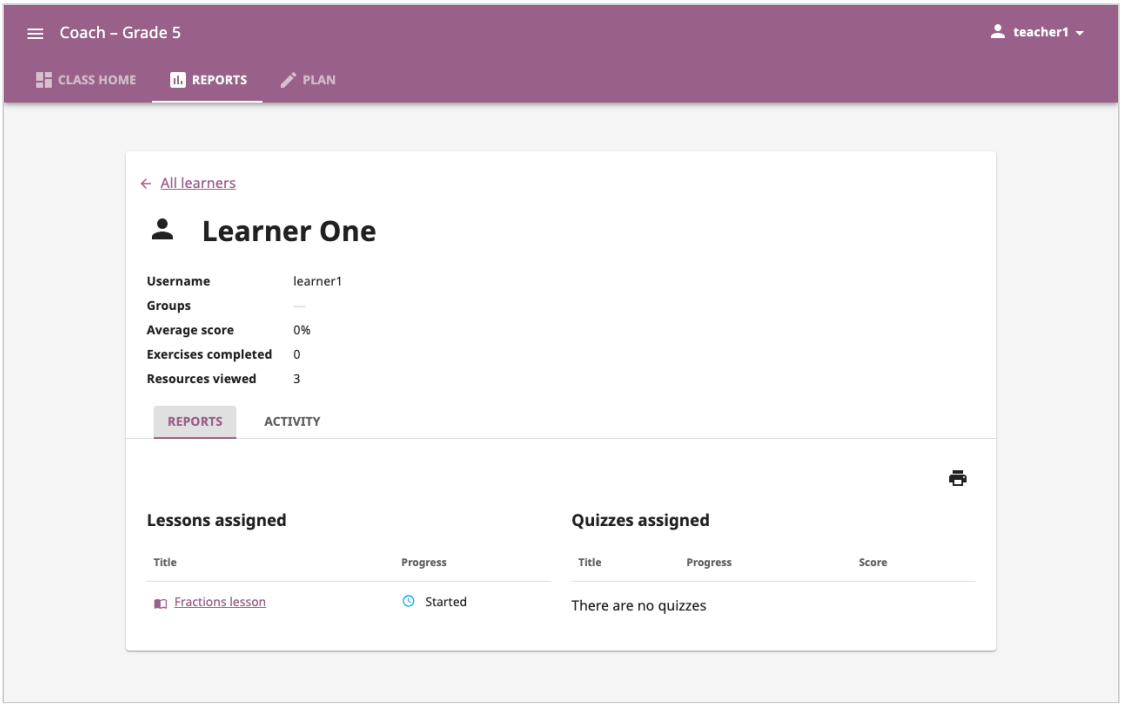


Figure 37. Learner report

Reports can be printed or exported as CSV (icons top right).

Tip. Do not chase 100%. Look for the question most learners got wrong and teach it again.

What the lesson report means

Kolibri shows	What it means	What to do
Completed	The learner finished the resource or the lesson	Nothing. Give them the next lesson or a quiz
Started	The learner began but did not finish	They can finish next time. Note the name
Has not started	The learner did not open it	Ask them. Often they did not sign in with their own name, or the lesson was not visible to learners

You do not need every learner at 100% after one class. Check that they signed in, worked through the explaining resources and tried the exercises.

3.7 Teaching one class period

This puts 3.4, 3.6 and chapter 4 together for one lesson. In Kolibri Demo, use `teacher1`, the class **Grade 5** and the **Maths lesson**.

Before class

1. **Do:** sign in as `teacher1`, open **Coach > Grade 5 > PLAN > LESSONS** and click **Maths lesson** (3.4).
   ✔ **Check:** the resources are in the order you want: explaining resources (videos, pictures, documents) first, exercises last.
2. **Do:** check **Recipients**.
   ✔ **Check:** it shows **Entire class** or the class **Grade 5**.
3. **Do:** switch on **Visible to learners**.
   ✔ **Check:** the switch is on. Learners see the lesson the next time they open **HOME**.

During class

1. **Do:** ask every learner to open Kolibri Demo and sign in with **their own** username (chapter 4).
   ✔ **Check:** each learner sees **Maths lesson** on **HOME**. A learner who uses someone else's name, or does not sign in, does not appear in Reports.
2. **Do:** tell the class to go through the explaining resources first, then do the exercises, in the order of the lesson.
3. **Do:** walk the room. Help learners who are stuck, but do not do the exercises for them.

After class

1. **Do:** open **REPORTS > LESSONS > Maths lesson** (3.6).
   ✔ **Check:** each learner is listed as **Completed**, **Started** or **Has not started**. The table in 3.6 says what to do about each.
2. **Do:** pick the one thing most learners found hard and teach it again next time. Do not chase 100%.

If something is wrong

What you see	What to do
The lesson is not on a learner's screen	Check that **Recipients** shows the class and that **Visible to learners** is on
A learner cannot sign in	Check that they type their own username. If it still does not work, tell your school operator
Reports is empty	Learners must sign in with their own name before they start
A resource is missing from the lesson	Tell your school operator
The Kolibri card in the Console has no **Open**	Tell your school operator. Do not try to fix it yourself

4. Kolibri for learners (try it yourself)

Your learners open Kolibri the same way you do: on a classroom computer they double-click **IDEA** on the desktop and click **Open** on the Kolibri card in the Console (2.2.1). Show them this once. To try it yourself with the demo learners, open **Kolibri Demo**.

Sign in as a learner in a second browser window (private window) to see exactly what your class sees. Learners have no password in Kolibri. They sign in with only their username, for example `learner1`.

> **On every IDEA box**, both **Kolibri** and **Kolibri Demo** already let learners sign in without a password. **At an existing school**, a teacher with admin rights (or the Kolibri admin) turns this on in Kolibri: open **Facility > SETTINGS**, untick **Require password for learners**, then click **SAVE CHANGES**.

4.1 Do a lesson

1. Sign in as `learner1`. **HOME** shows **Your classes**, **Recent lessons** and **Recent quizzes**.

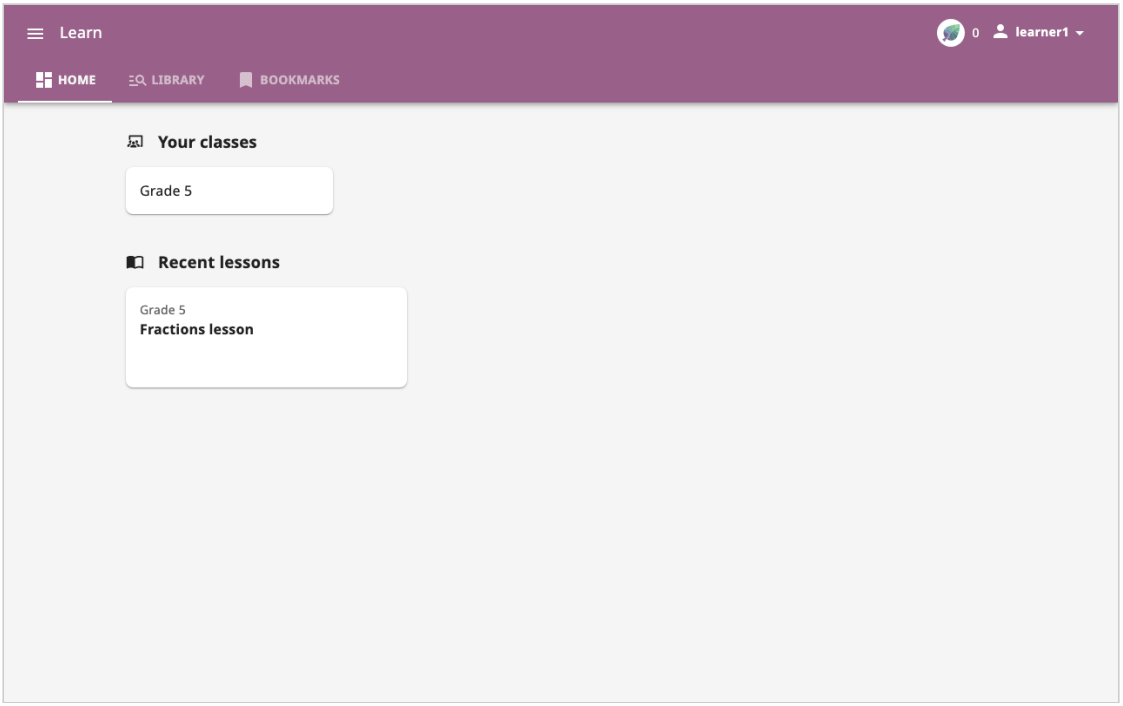


Figure 38. Learner home

2. Click **Fractions lesson**. Its items are listed in order.

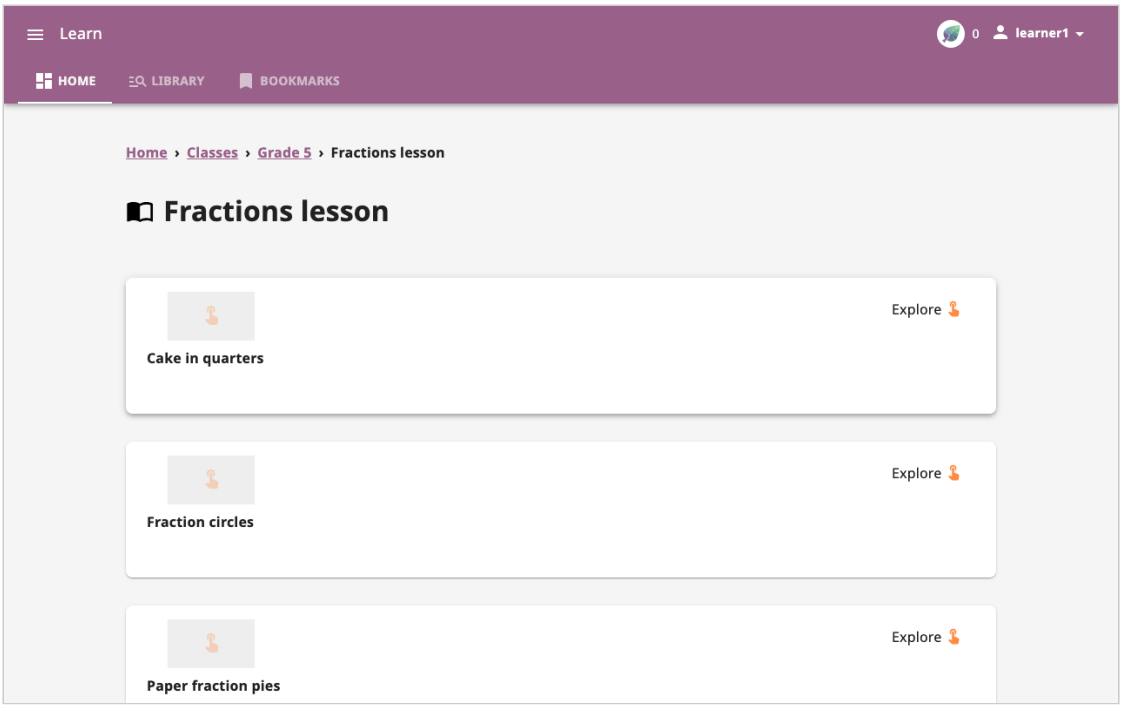


Figure 39. Lesson

3. Open the first item (**Cake in quarters**, a picture activity). Videos open the same way.

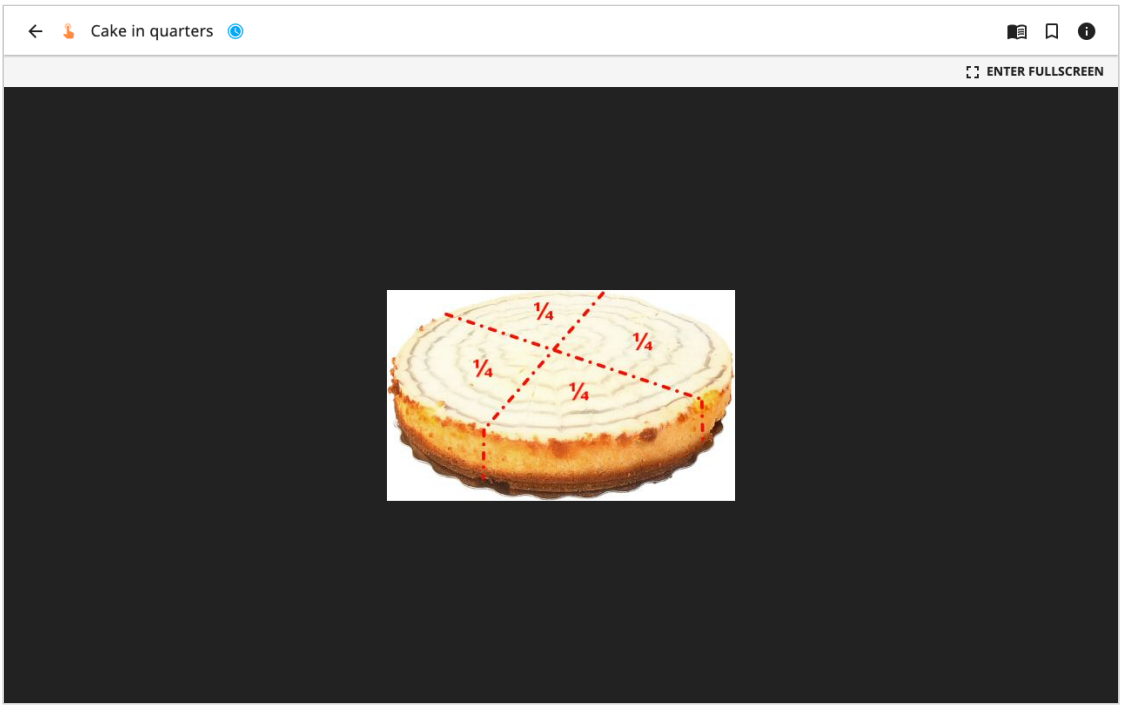


Figure 40. Video

4. Exercises work like this (here a Khan Academy exercise; in Kolibri Demo, try **Fractions practice** in the **Maths lesson**, see 3.5): type an answer, click **CHECK**. Keep going until the goal at the top ("Get 5 correct") is reached.

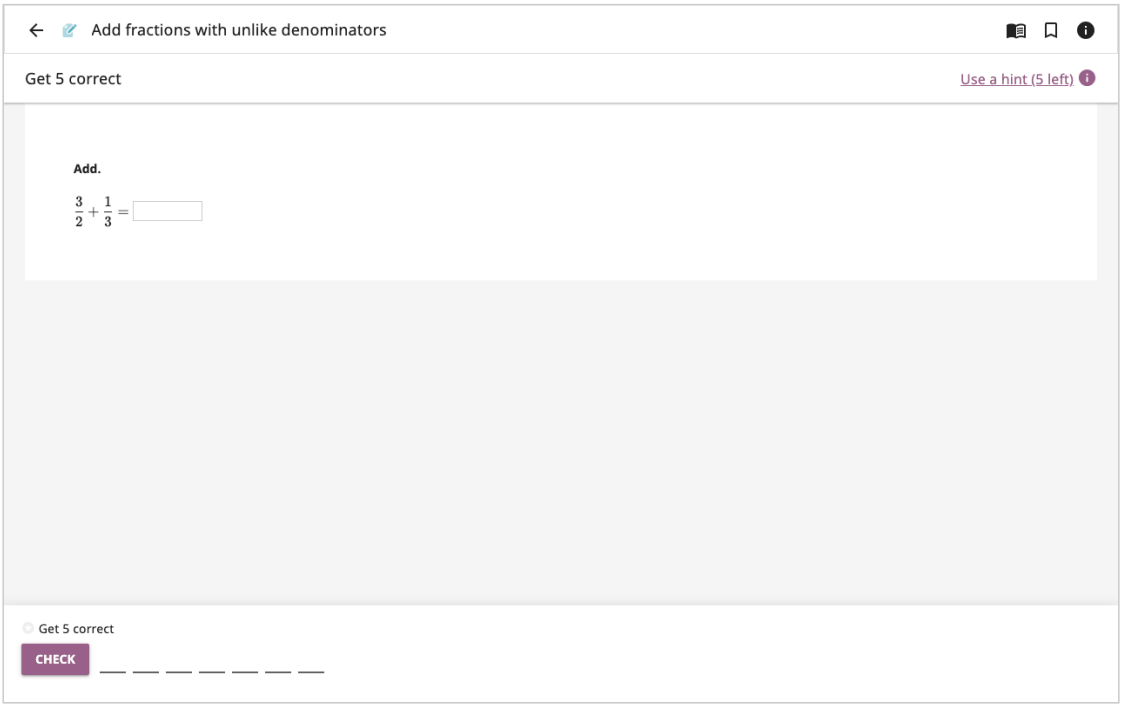


Figure 41. Exercise

5. Sign out. Sign in as `learner2` and open only one item, so the report shows a difference.

4.2 Take a quiz

This needs a quiz that has been started. In Kolibri Demo, use **Grade 5 Practice Quiz** (see 3.5).

1. Open the quiz from **Recent quizzes** (or the class page).

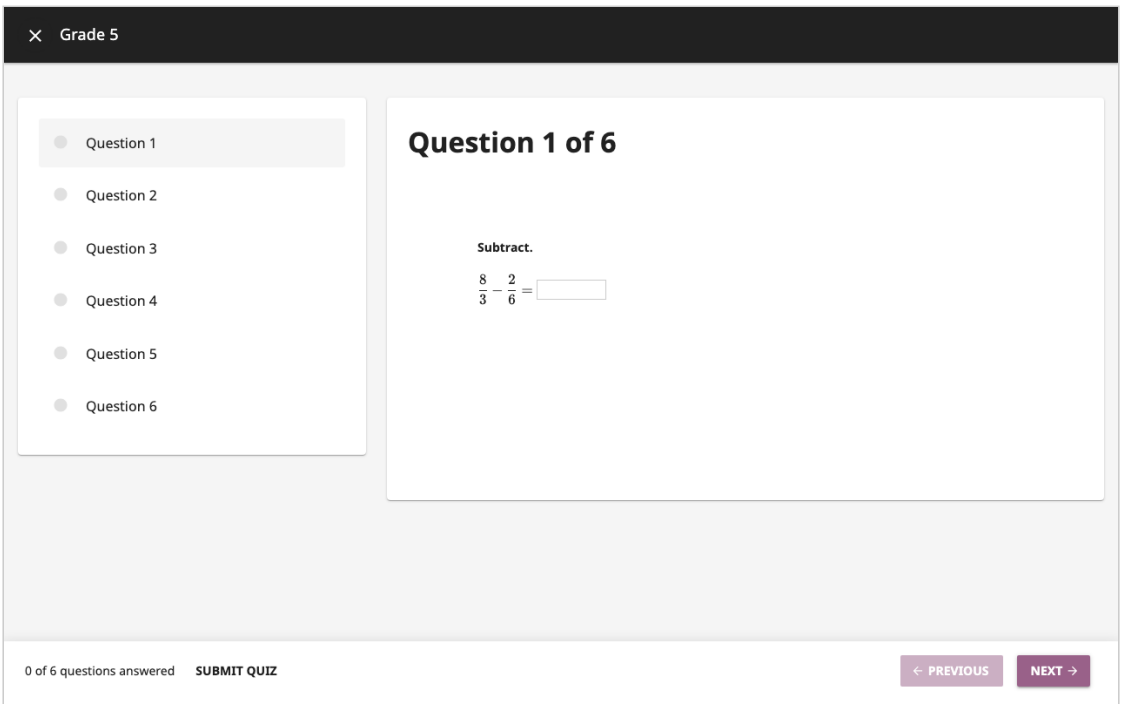


Figure 42. Quiz

2. Answer each question and use **NEXT** / **PREVIOUS**. The list on the left shows answered questions.

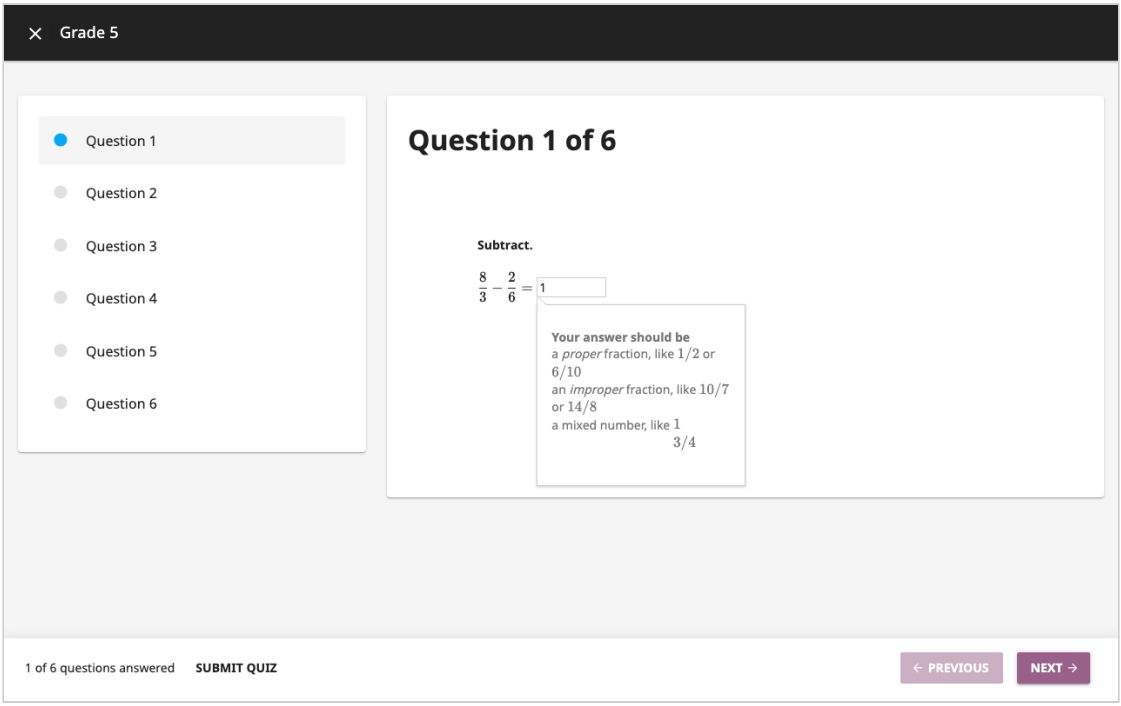


Figure 43. Question

3. Click **SUBMIT QUIZ** and confirm. After submitting, answers cannot be changed.

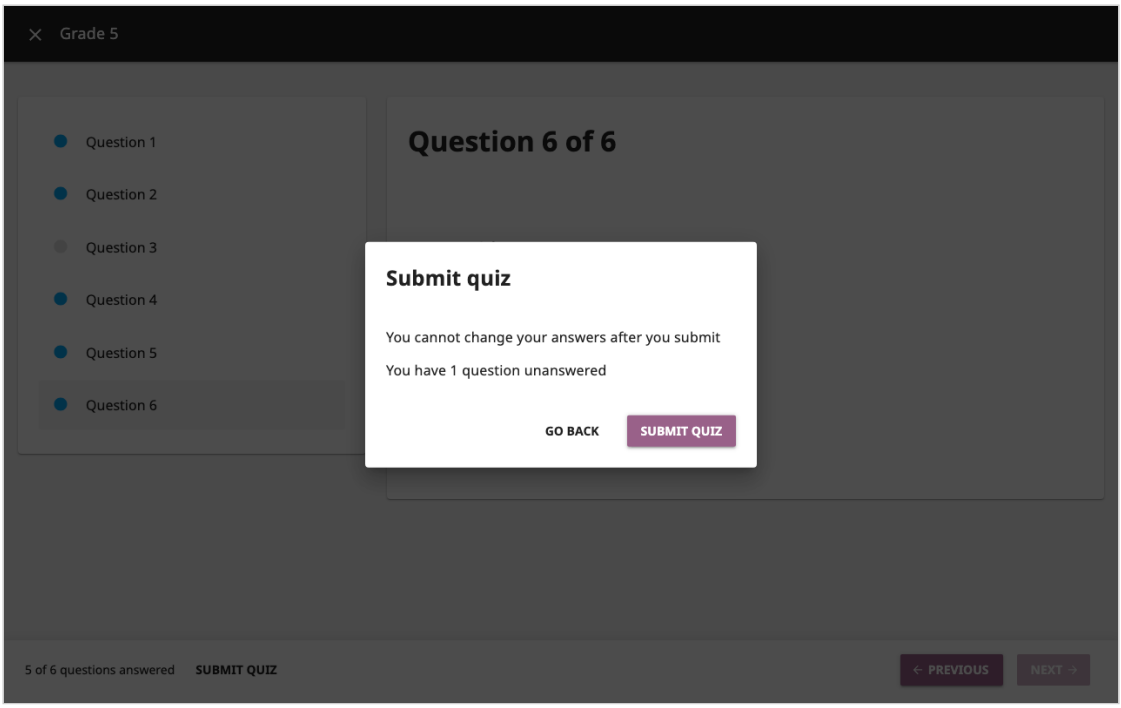


Figure 44. Submit

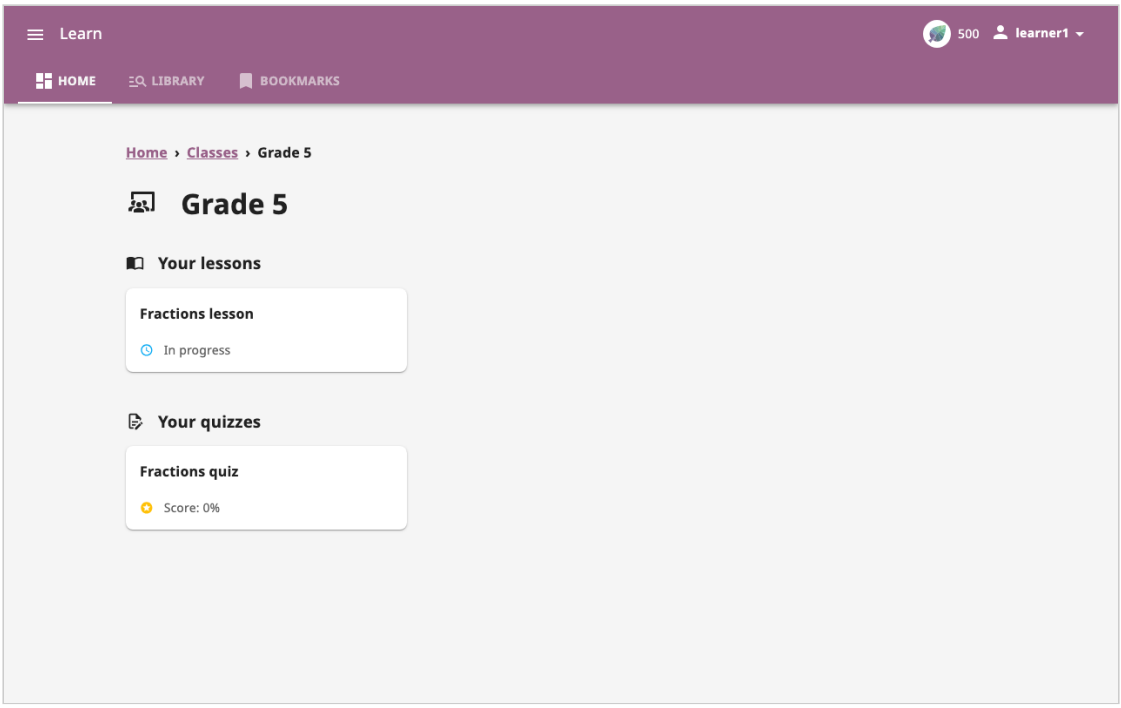


Figure 45. After submit

Now go back to the teacher window and look at Reports (3.6).

5. Nextcloud for teachers

About the examples in this chapter. Every IDEA box comes with the demo setup switched on: the app **Nextcloud Demo**. It has the demo teachers `teacher1` and `teacher2` (group `teachers`), the demo learners `learner1` to `learner3` (group `learners`), and in each teacher's files the folders `Worksheets` and `Teacher Notes`. The screens in this chapter call the class group `Grade 5`: in Nextcloud Demo, pick `learners` wherever the steps say the class group. The steps work the same at your school, but your names will be different. The demo accounts are listed in Appendix: demo accounts.

Office documents open in **Nextcloud Office**.

Open **Nextcloud Demo** from the Console (chapter 2) and sign in as `teacher1`. Click **Files** at the top.

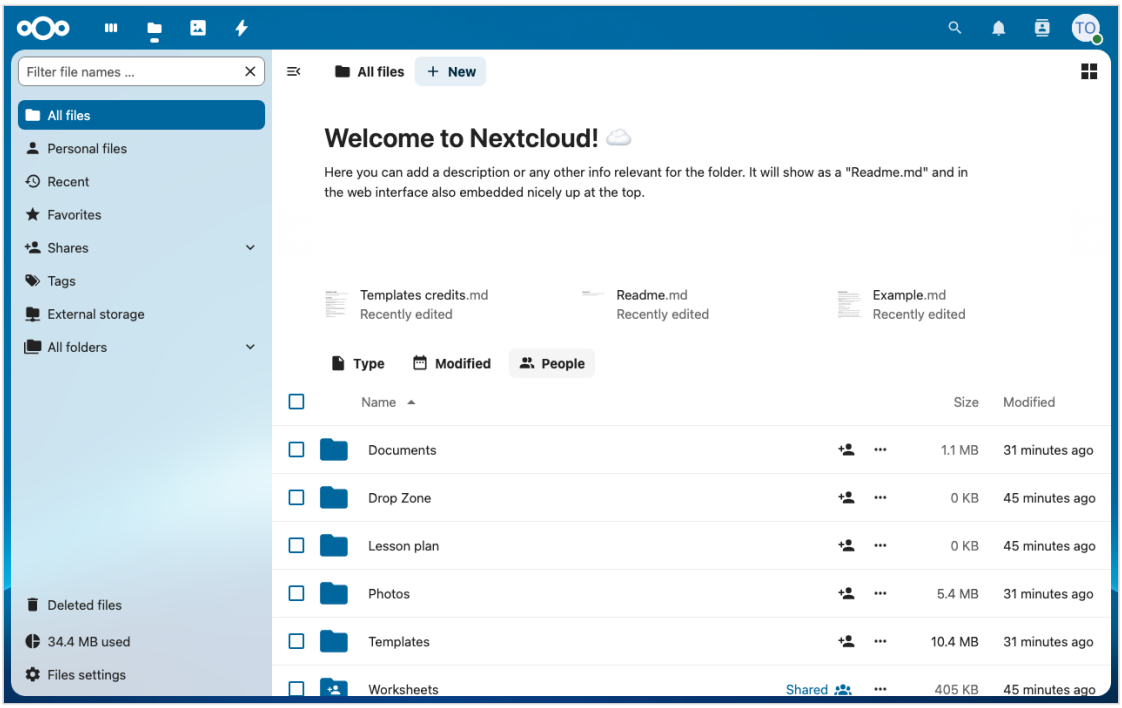


Figure 46. Files

5.1 Share worksheets with the class (view only)

1. Next to the folder `Worksheets`, click the share icon (**Show sharing options**). The right panel opens on **Sharing**.

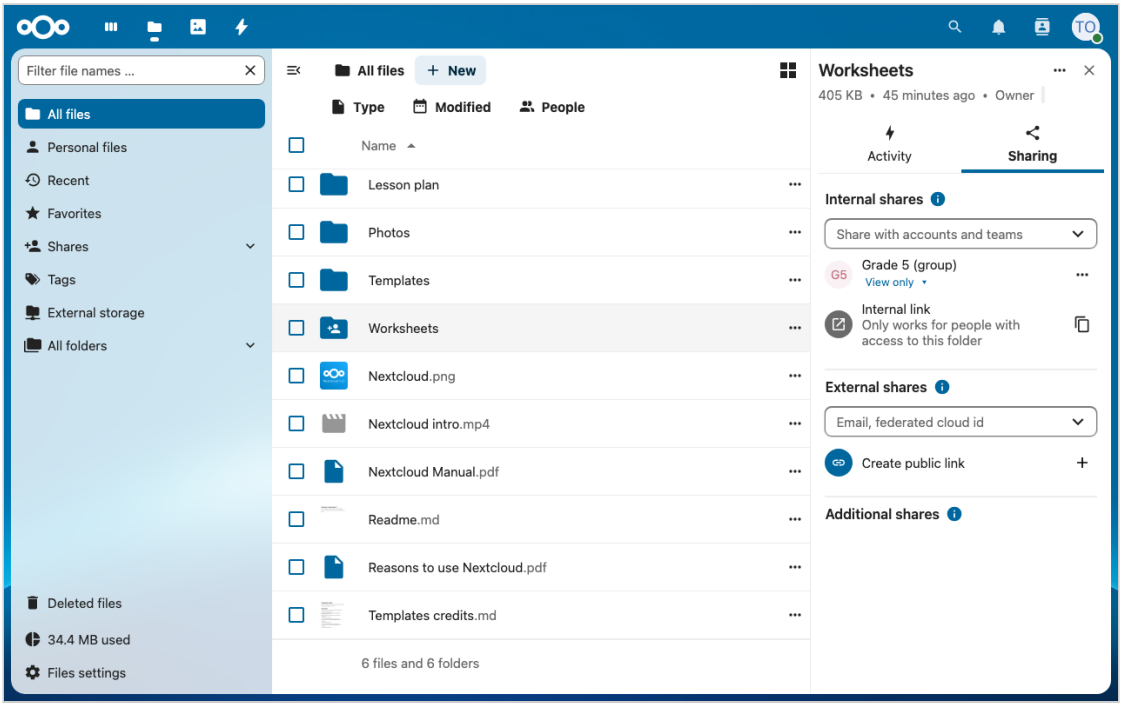


Figure 47. Sharing panel

2. In **Internal shares** ("Share with accounts and teams"), type the class group (on the kit: `learners`) and pick the group.
3. Set the permission to **View only**. The group then appears under Internal shares.

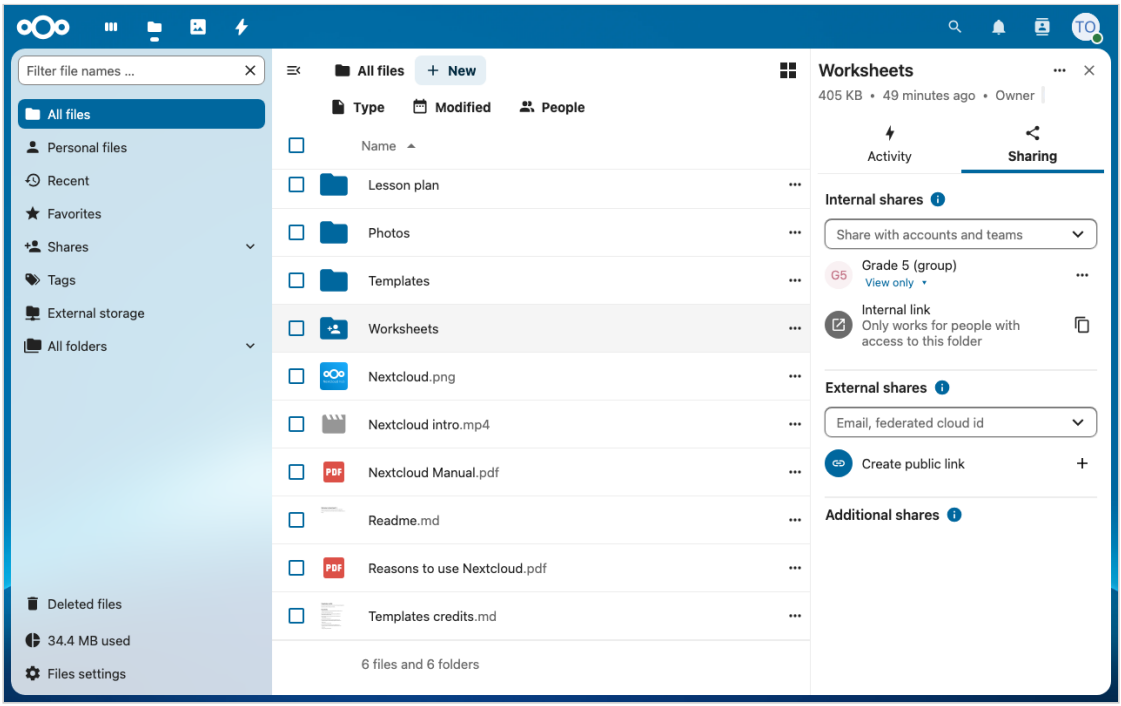


Figure 48. Shared with group

Learners now see the folder but cannot change or delete files.

5.2 Write a document together

1. Click **+ New > New document**.

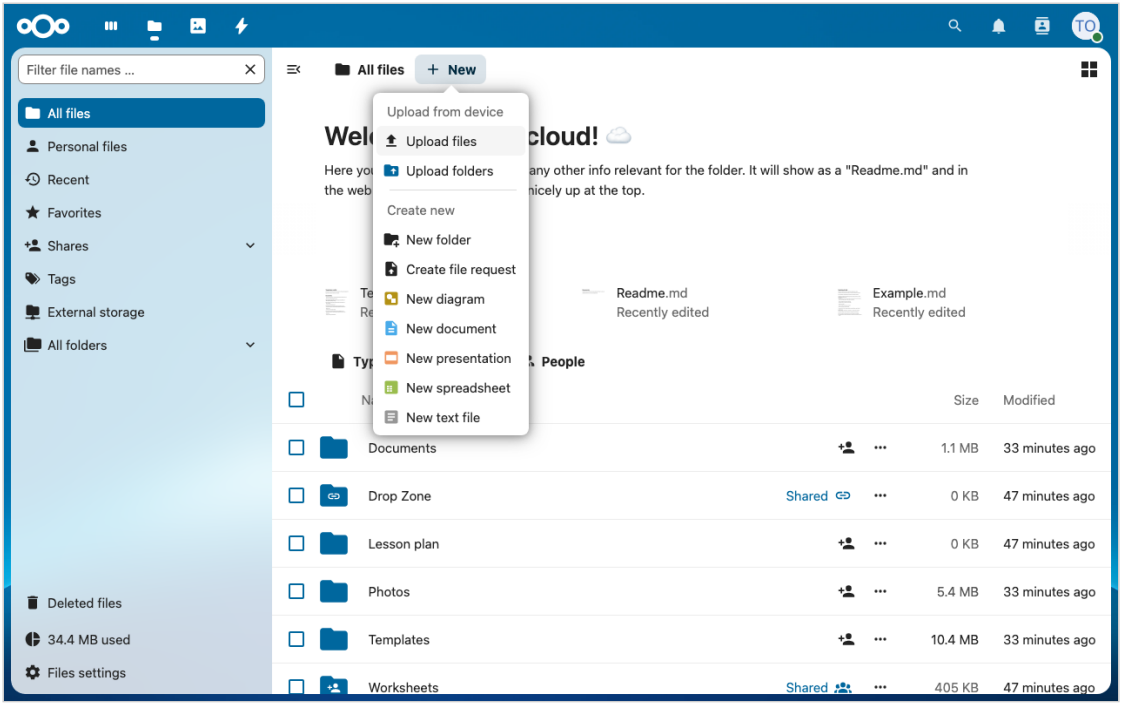


Figure 49. New menu

2. Type a name (`Lesson plan`) and click **Create**.

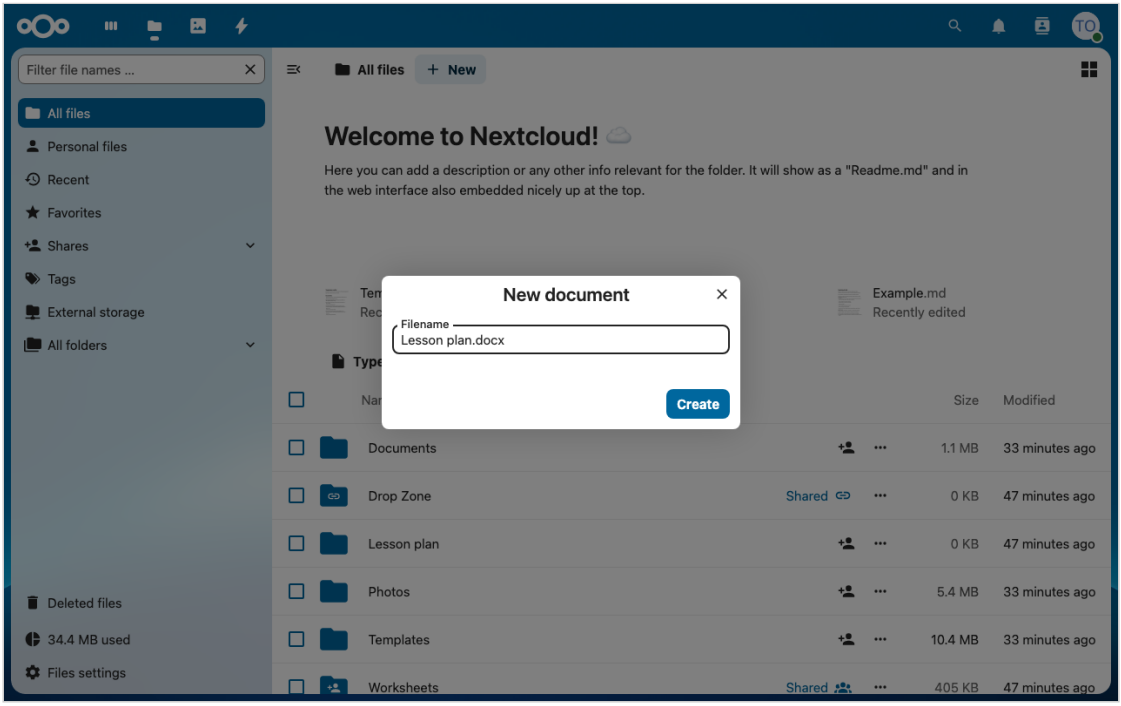


Figure 50. Name

3. Pick a template (Blank is fine) and click **Create**.

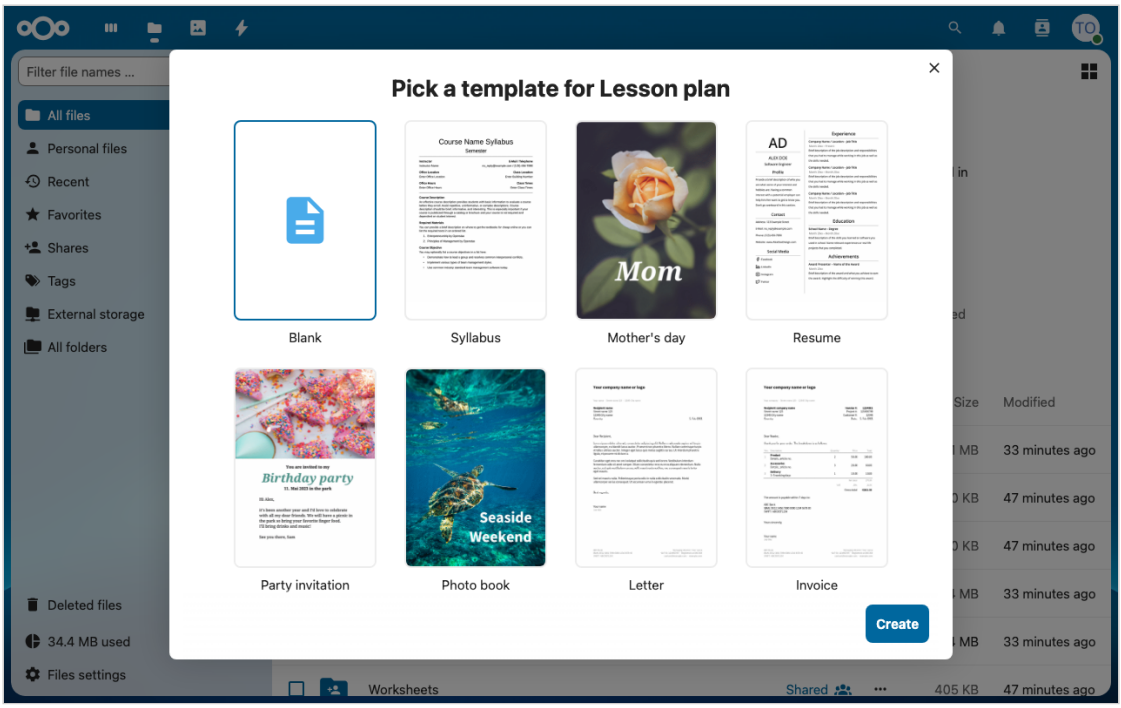


Figure 51. Template

4. Share the file with `teachers` and/or your class group (on the kit: `learners`), with editing allowed.

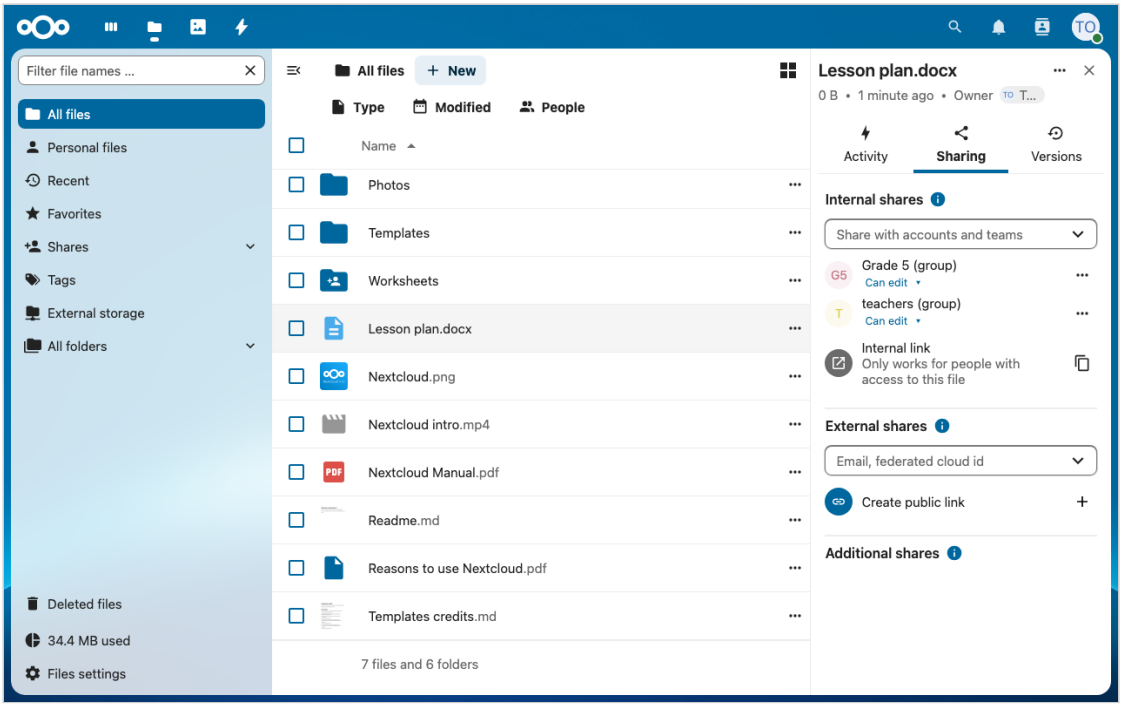


Figure 52. Shared document

5. The document opens in Nextcloud Office. When others open it at the same time, you see their names and cursors.

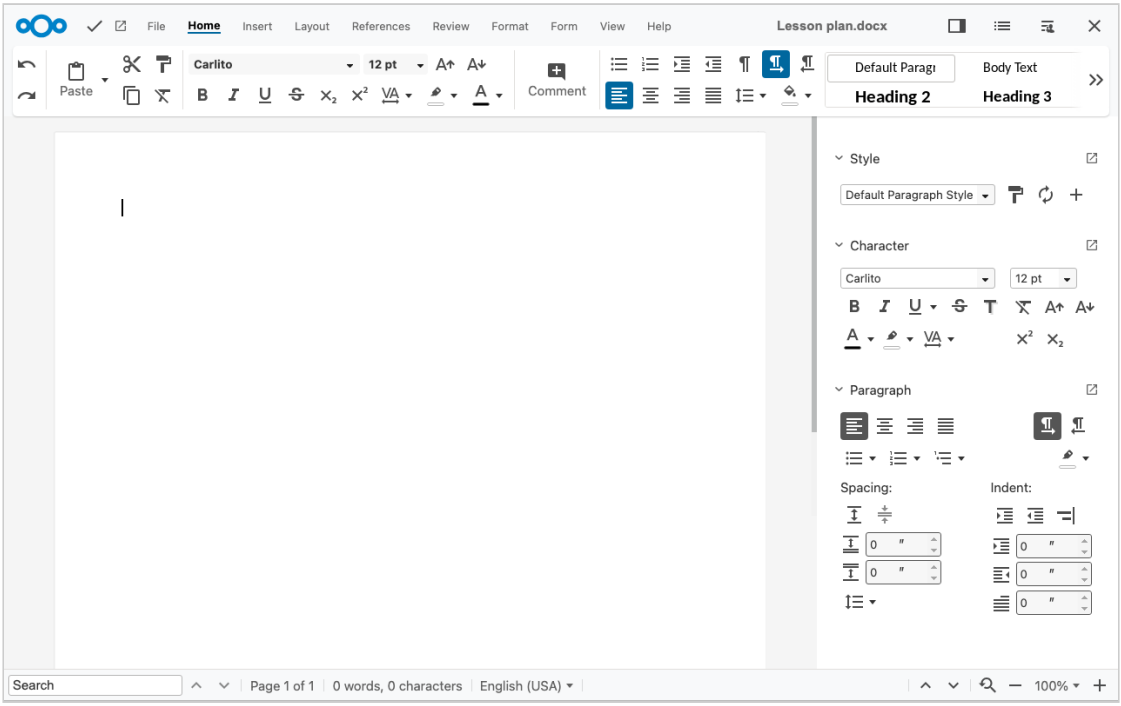


Figure 53. Nextcloud Office

5.3 Collect homework (File drop)

Learners upload work, but cannot see each other's work.

1. Make a folder for the homework: **+ New > New folder**, name it `Drop Zone`. Then open sharing for the folder `Drop Zone`.

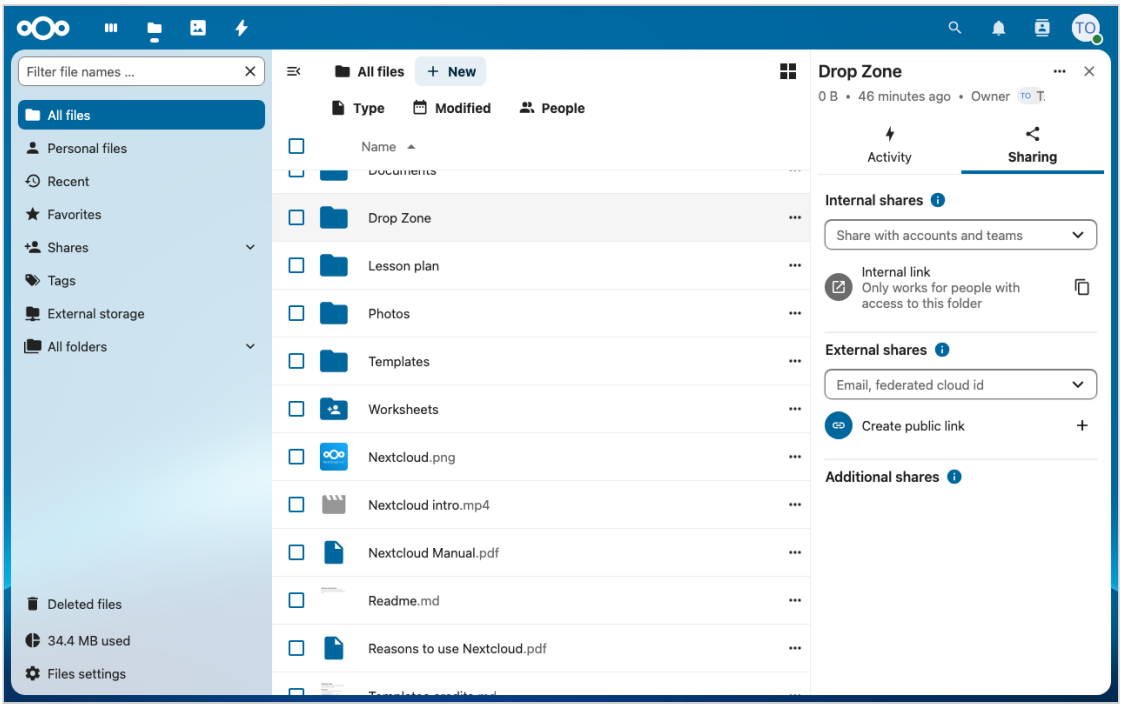


Figure 54. Sharing for Drop Zone

2. Under **External shares**, click **+** next to **Create public link** and choose **File request** (upload only). Copy the link.

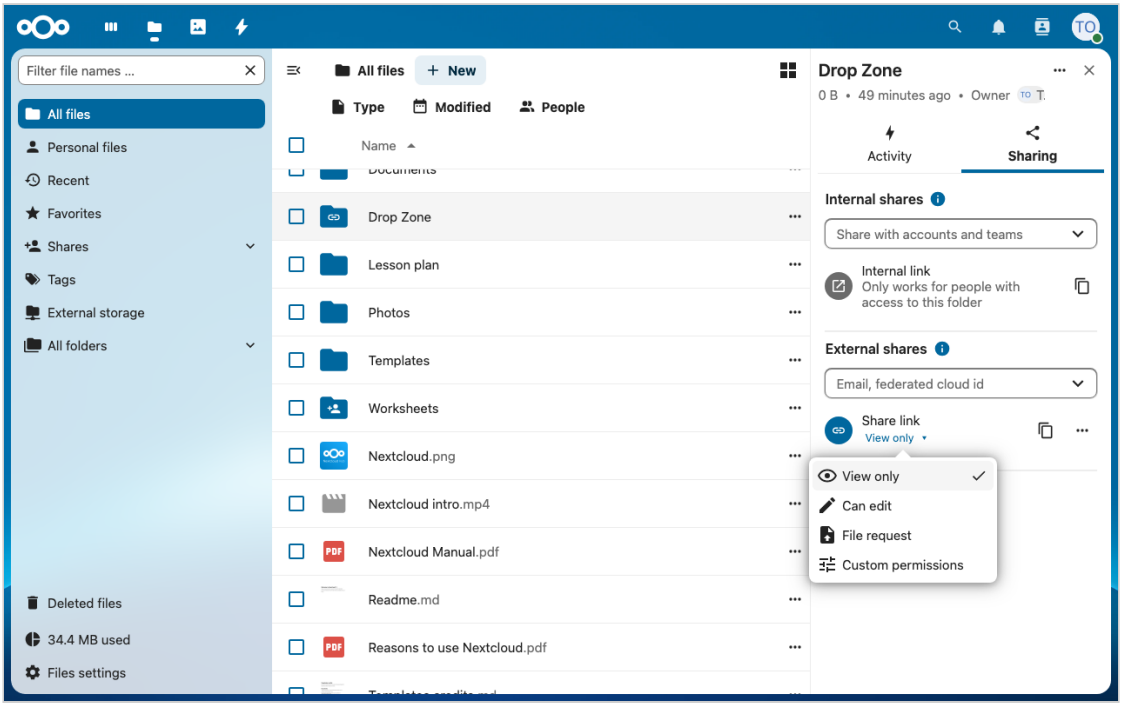


Figure 55. Create link

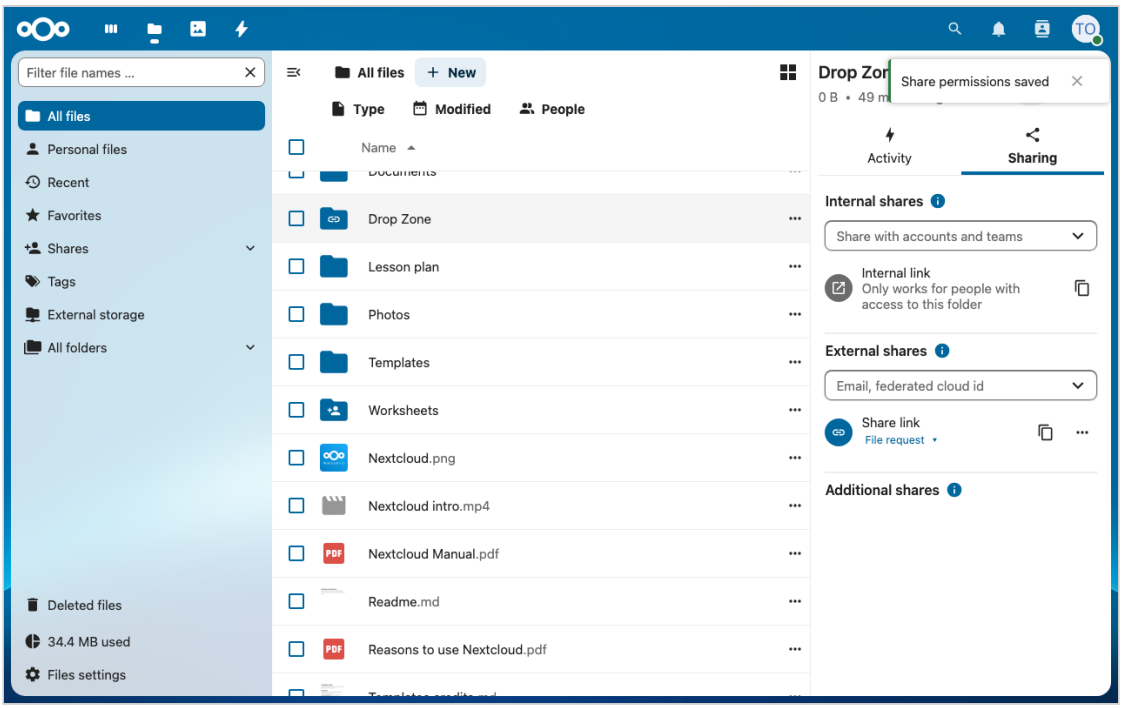


Figure 56. Link created

3. Give learners the link (for example, write it on the board). Uploaded files appear in your `Drop Zone` folder.

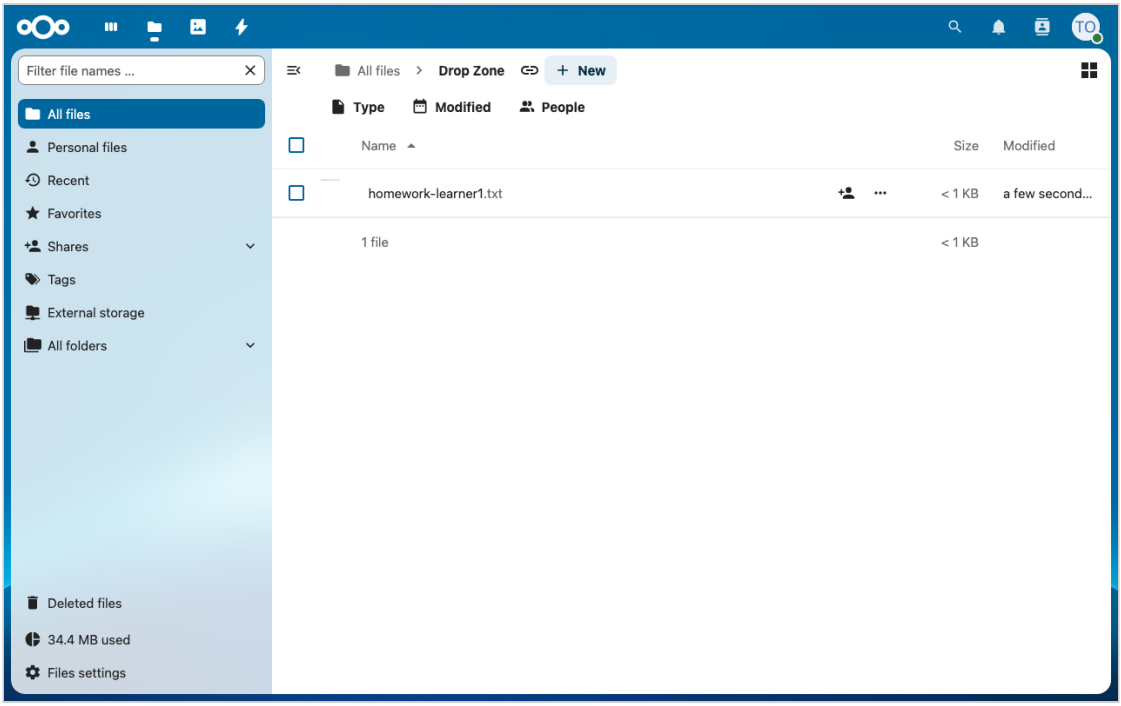


Figure 57. Received homework

6. Nextcloud for learners (try it yourself)

Learners open Nextcloud from the Console, like Kolibri: **IDEA** on the desktop, then **Open** on the Nextcloud card (2.2.1). They have no personal Nextcloud app on the classroom computer, so they always work in the browser.

6.1 Open class files and co-edit

1. Sign in to **Nextcloud Demo** as `learner1`, password `learner1911!` (private window). Shared folders and files appear in **Files** with the teacher's name.

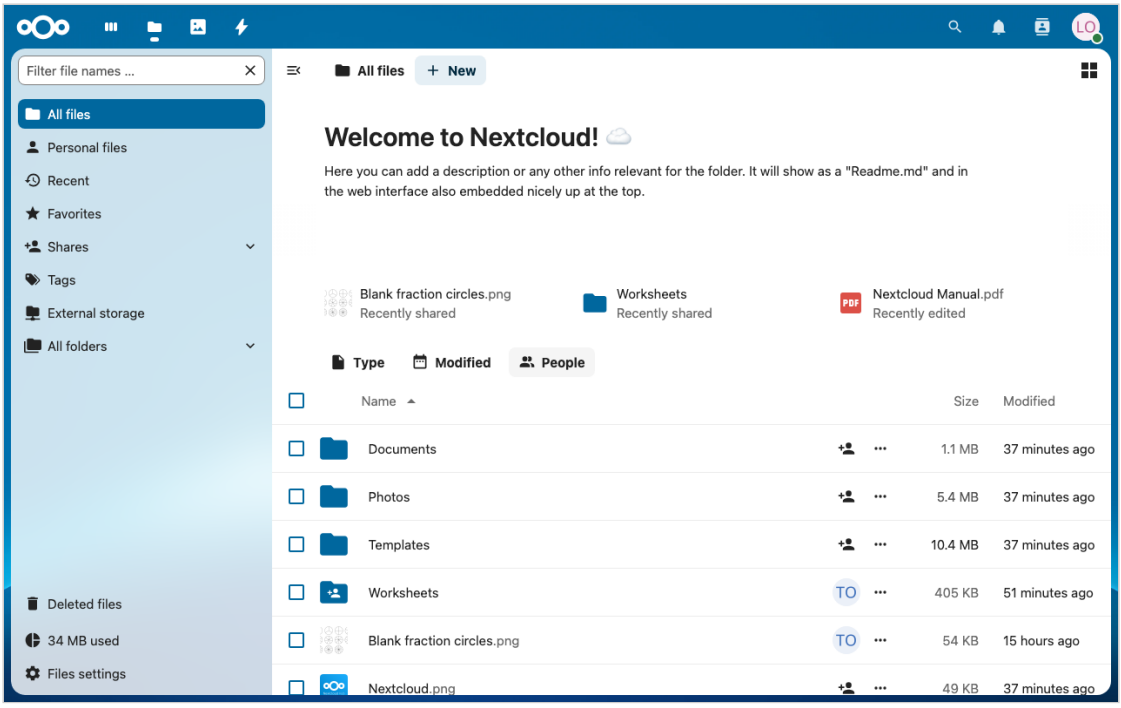


Figure 58. Learner files

2. Open `Worksheets`: the learner can open and download the worksheets, not change them.

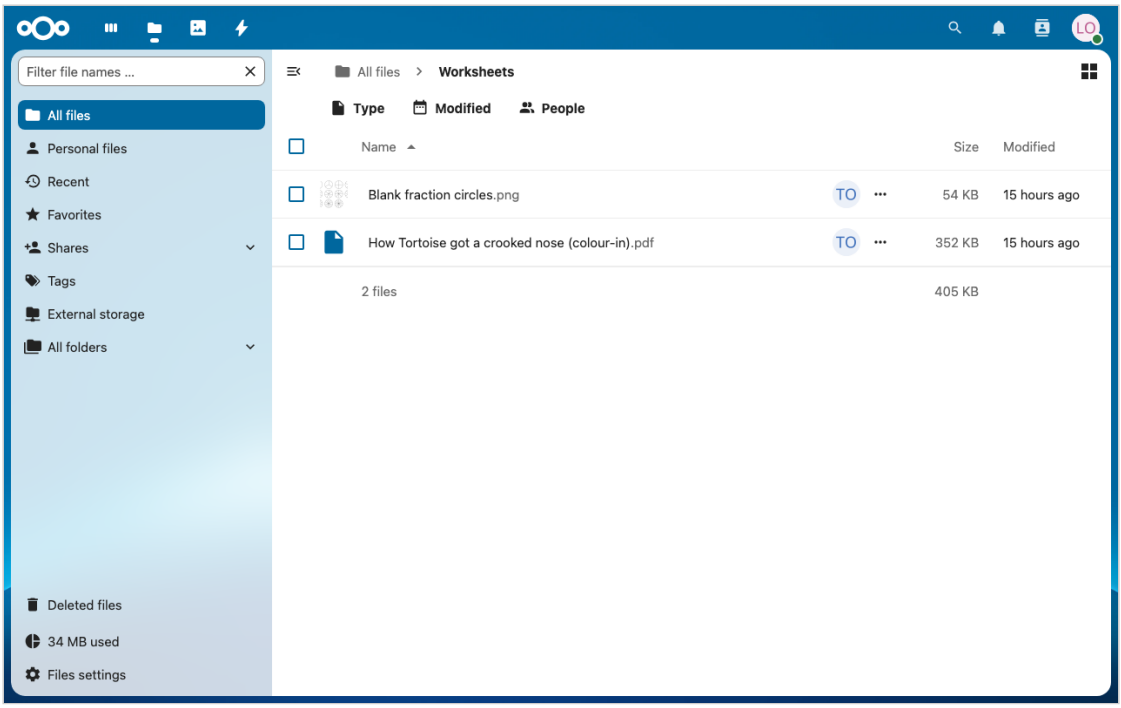


Figure 59. Worksheets

3. Open `Lesson plan.docx`: it opens in Nextcloud Office and the learner can type.

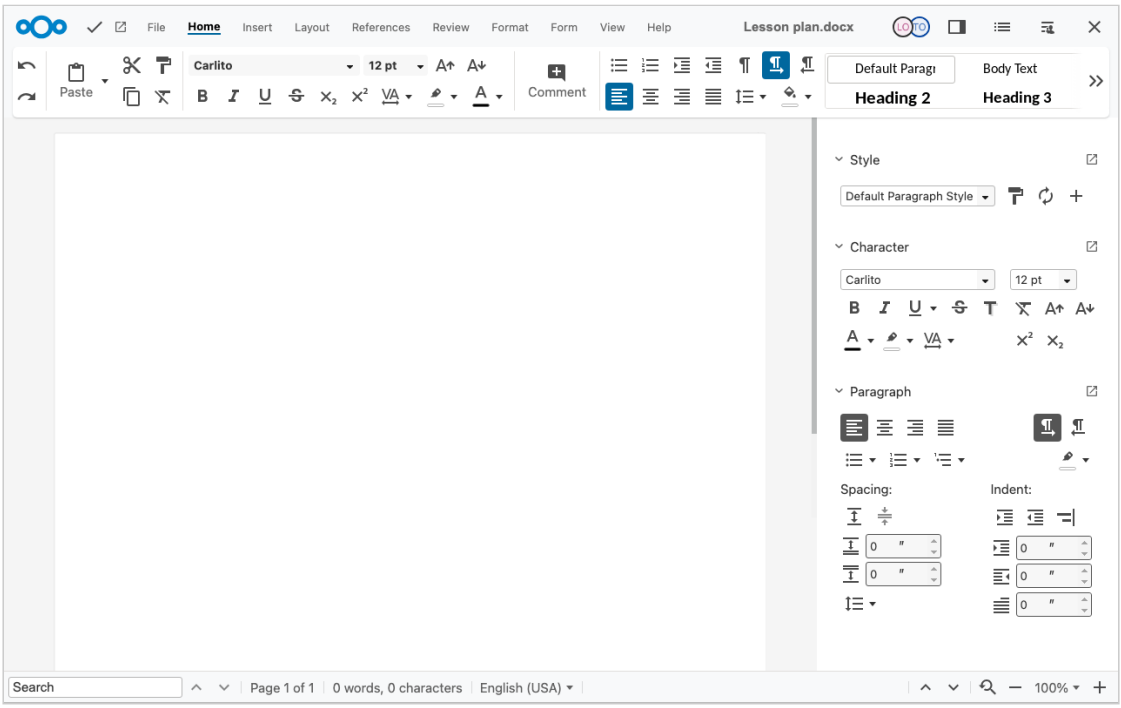


Figure 60. Learner in document

6.2 Hand in homework

1. Open the File drop link from 6.3. The page says **File drop** and shows an **Upload** button; the learner cannot see other files.

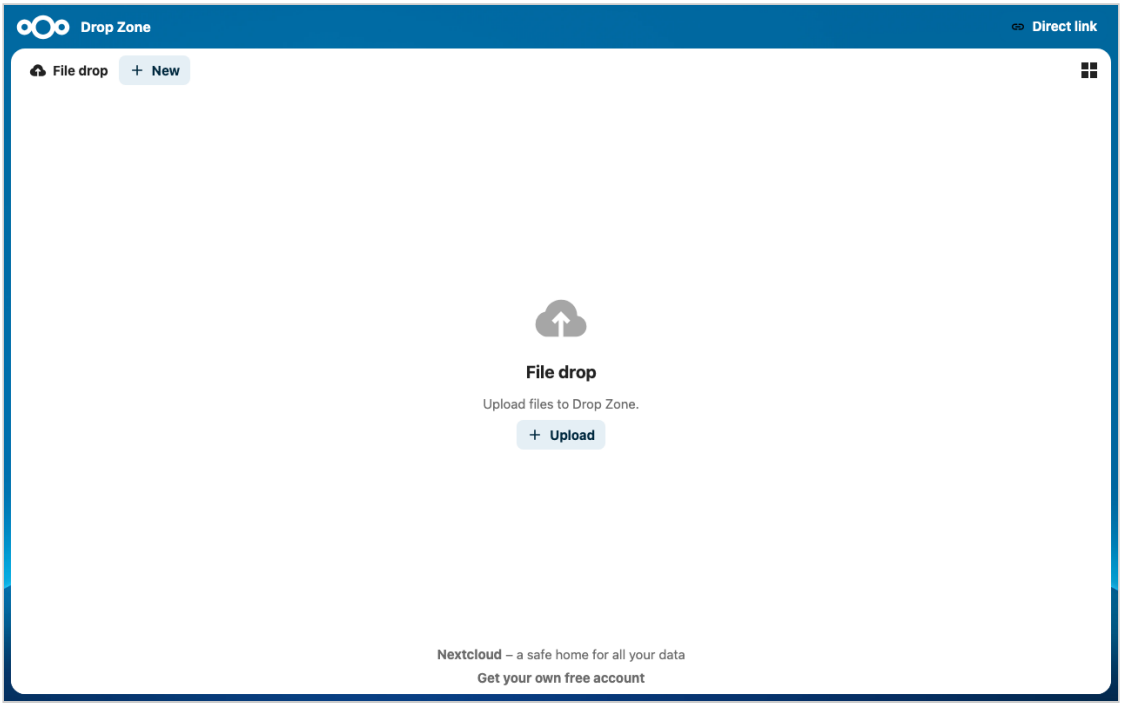


Figure 61. File drop

2. Click **Upload** and choose the file (e.g. a photo of written work). If asked, type your name.

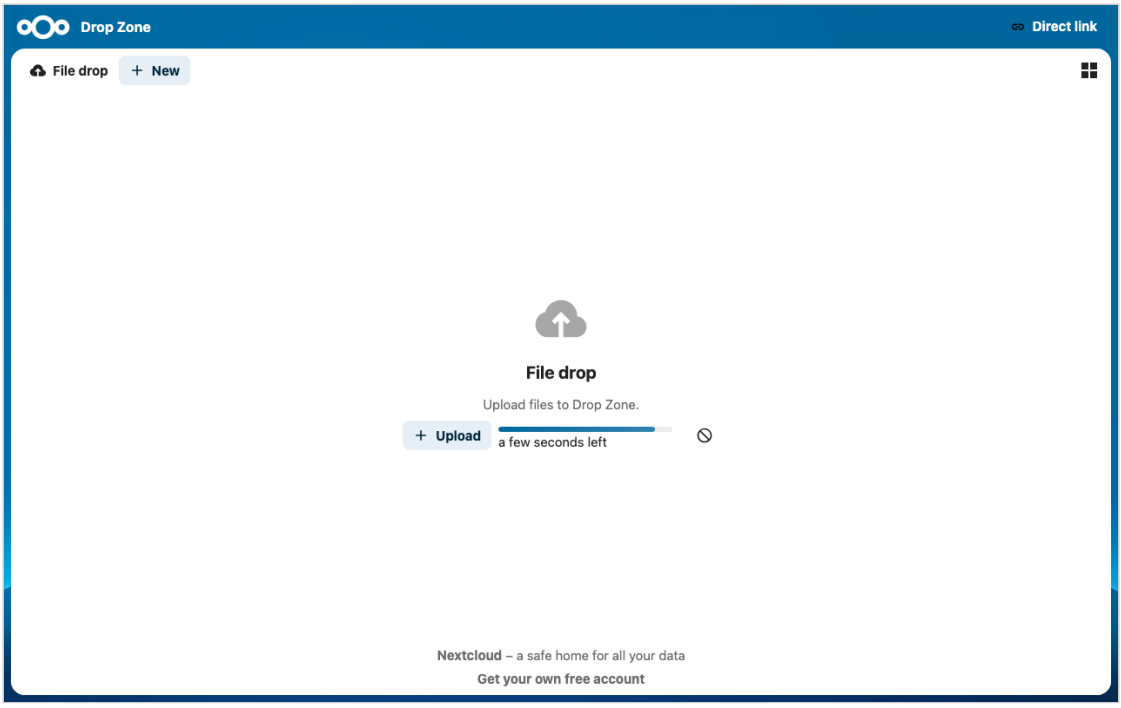


Figure 62. Uploading: the bar shows the progress. When it is gone, the file is in the teacher's folder

The teacher sees the file in `Drop Zone` (5.3).

7. Wikipedia offline

Every IDEA box of the kit has two Wikipedia apps. Both work without internet.

Console card	What it has
Wikipedia	The full English Wikipedia.
Wikipedia (Demo)	**Wikipedia in simple English**: shorter articles in easier English. The steps below use it.

> **Two cards called Wikipedia.** On a computer screen the Console's **Apps** page can show both cards with the title **Wikipedia**. Open one: the library page names the book, and **Wikipedia in simple English** is the Demo.

7.1 Look something up and use it in class

1. Open **Wikipedia (Demo)** from the Console (chapter 2). The library page shows the book.

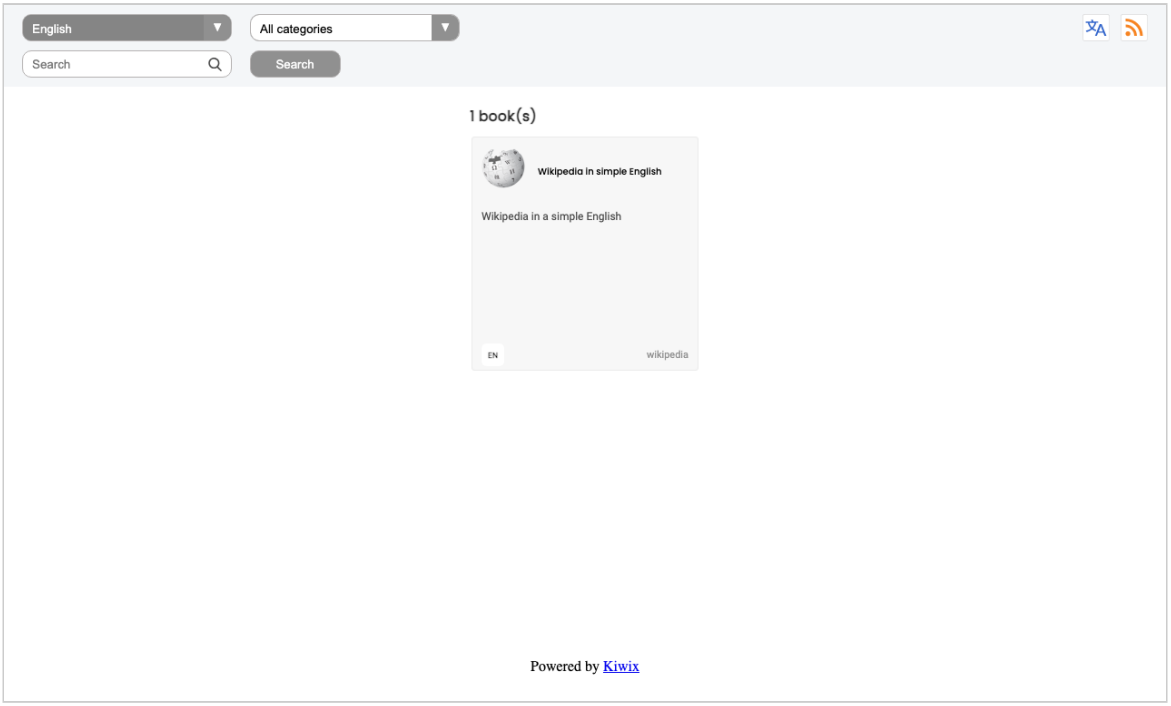


Figure 63. The library page

2. Click **Wikipedia in simple English**. Its welcome page opens.



Figure 64. Wikipedia home

3. Type in the search box at the top, e.g. `Fraction`. Pick a result.

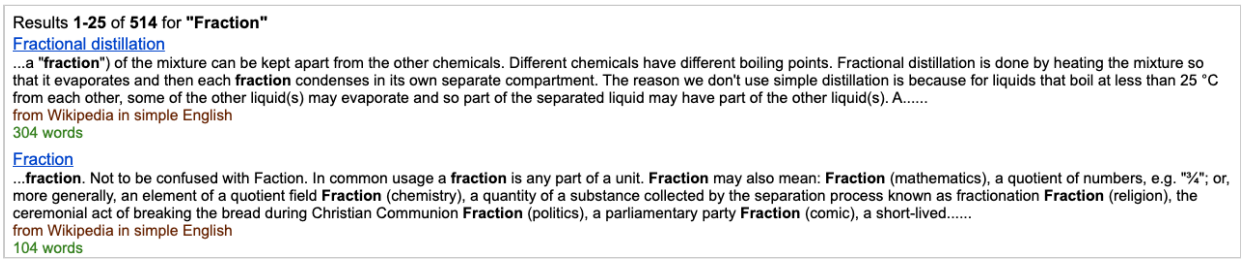


Figure 65. Search results

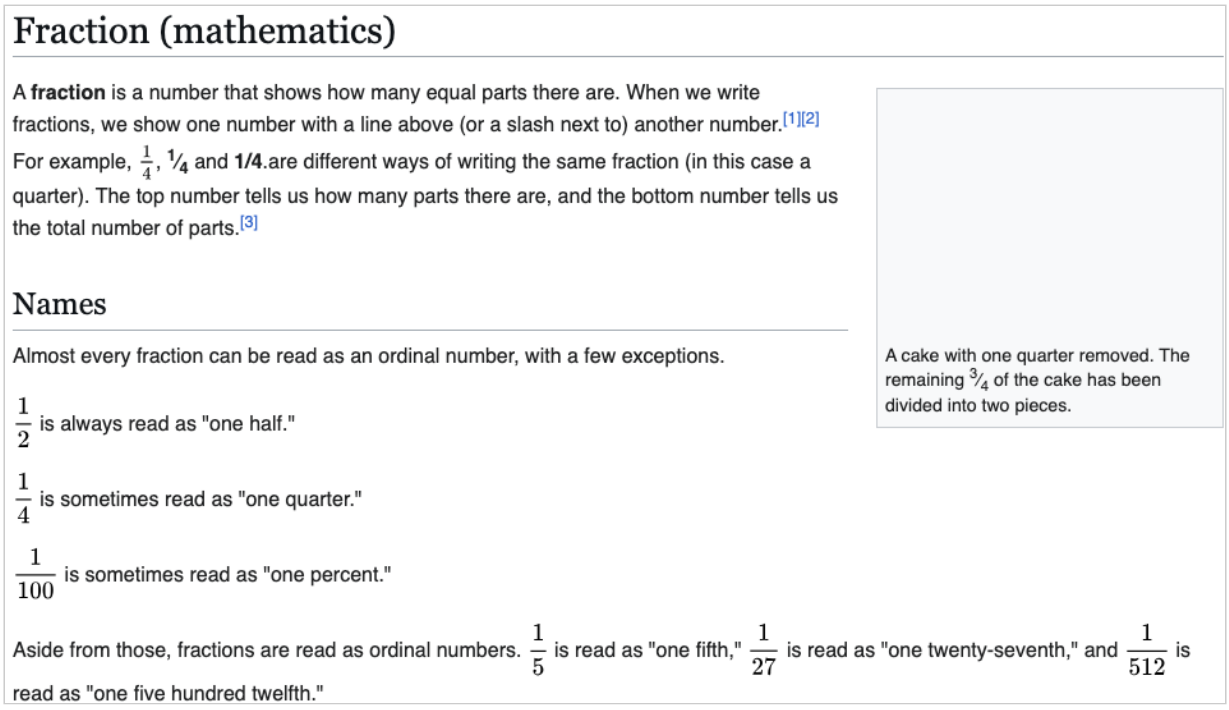
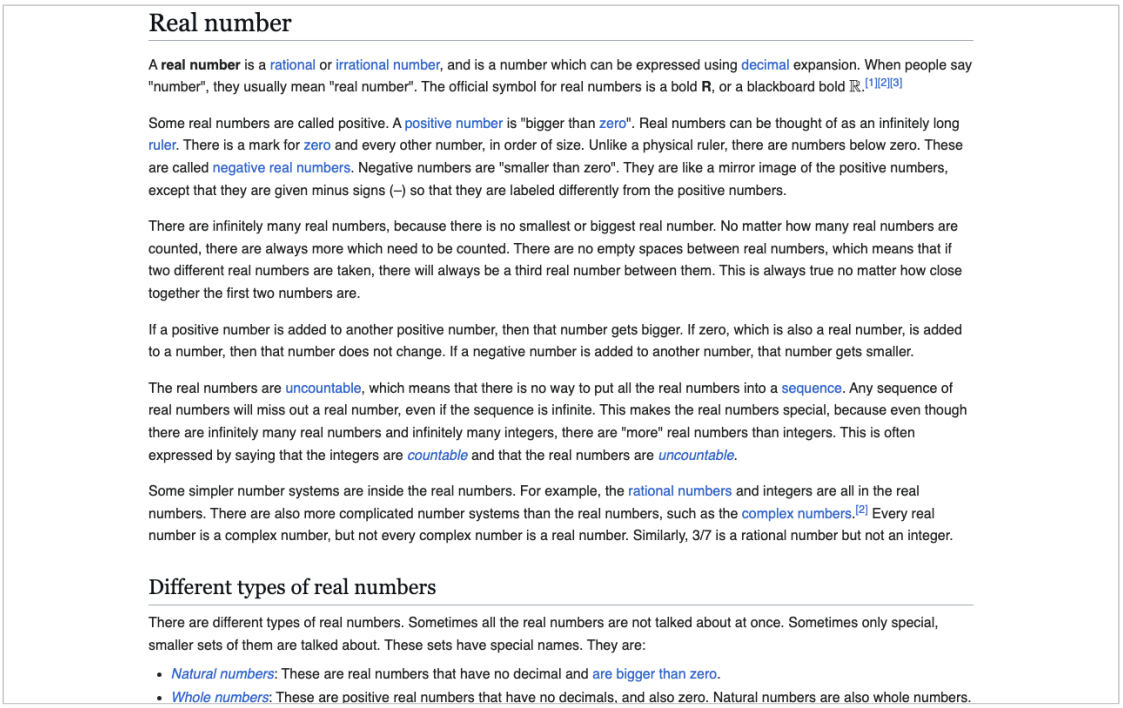


Figure 66. Fraction article

4. Follow blue links to related articles.



Real number

A **real number** is a rational or irrational number, and is a number which can be expressed using decimal expansion. When people say "number", they usually mean "real number". The official symbol for real numbers is a bold **R**, or a blackboard bold $\mathbb{R}$.[1][2][3]

Some real numbers are called positive. A positive number is "bigger than zero". Real numbers can be thought of as an infinitely long ruler. There is a mark for zero and every other number, in order of size. Unlike a physical ruler, there are numbers below zero. These are called negative real numbers. Negative numbers are "smaller than zero". They are like a mirror image of the positive numbers, except that they are given minus signs (–) so that they are labeled differently from the positive numbers.

There are infinitely many real numbers, because there is no smallest or biggest real number. No matter how many real numbers are counted, there are always more which need to be counted. There are no empty spaces between real numbers, which means that if two different real numbers are taken, there will always be a third real number between them. This is always true no matter how close together the first two numbers are.

If a positive number is added to another positive number, then that number gets bigger. If zero, which is also a real number, is added to a number, then that number does not change. If a negative number is added to another number, that number gets smaller.

The real numbers are uncountable, which means that there is no way to put all the real numbers into a sequence. Any sequence of real numbers will miss out a real number, even if the sequence is infinite. This makes the real numbers special, because even though there are infinitely many real numbers and infinitely many integers, there are "more" real numbers than integers. This is often expressed by saying that the integers are *countable* and that the real numbers are *uncountable*.

Some simpler number systems are inside the real numbers. For example, the rational numbers and integers are all in the real numbers. There are also more complicated number systems than the real numbers, such as the complex numbers.[2] Every real number is a complex number, but not every complex number is a real number. Similarly, 3/7 is a rational number but not an integer.

Different types of real numbers

There are different types of real numbers. Sometimes all the real numbers are not talked about at once. Sometimes only special, smaller sets of them are talked about. These sets have special names. They are:

- *Natural numbers*: These are real numbers that have no decimal and are bigger than zero.
- *Whole numbers*: These are positive real numbers that have no decimals, and also zero. Natural numbers are also whole numbers.

Figure 67. Linked article

> **Classroom idea.** Give learners 3 questions to answer from one article; they read on their own device.

8. Your own files: the Files Disk

A **Files Disk** is an **IDEA Disk** that the operator prepares in the Console (IDEA for Operators: Tutorial). Its content appears in Nextcloud as the folder **School Files**. Use it to bring in material you downloaded elsewhere, for example worksheets you copied at another school. On the kit, every IDEA box has its own Files Disk.

Good to know:

- **School Files** is only in **Nextcloud**, not in Kolibri.
- Only teachers see the **School Files** folder. Learners get only the files you share with them.
- **The Backup Disk does not save School Files.** Keep your own copy of important files.

8.1 Open School Files

1. Open Nextcloud from the Console and sign in. In **Files** you see the folder **School Files** (marked as **External storage** in the left menu).

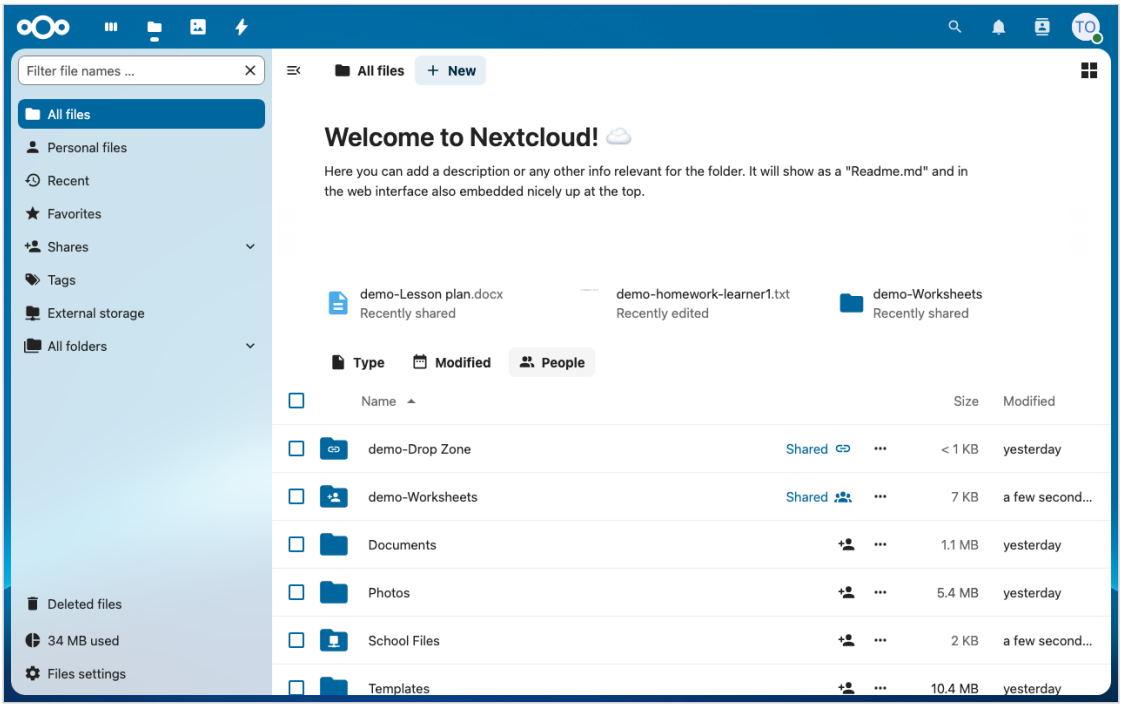


Figure 68. School Files in Nextcloud

2. Open **School Files** to see what is on the disk (here `Fractions-Worksheet-1.pdf`).

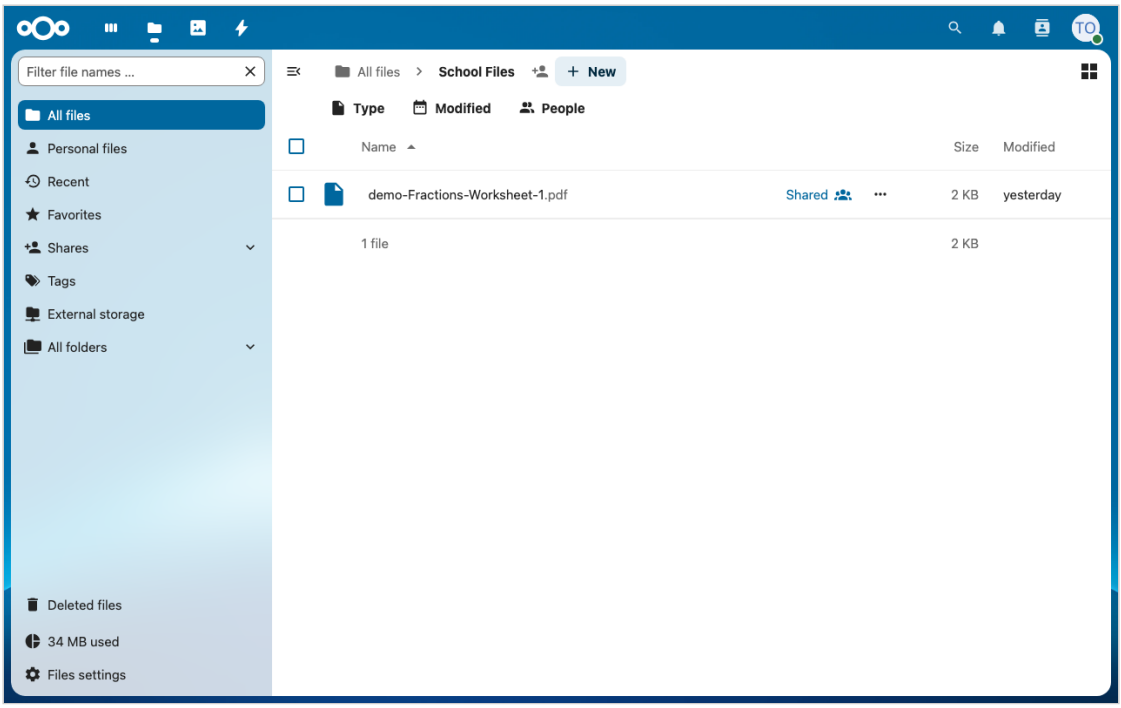


Figure 69. School Files content

8.2 Share a file from School Files with your class

1. On the file's row click **··· (Actions) > Show sharing options**, or click the share icon.
2. In **Internal shares**, type the class group name (here `Grade 5`; in Nextcloud Demo on the kit, `learners`) and choose it from the list. Keep **View only**. The group then appears under **Internal shares**.

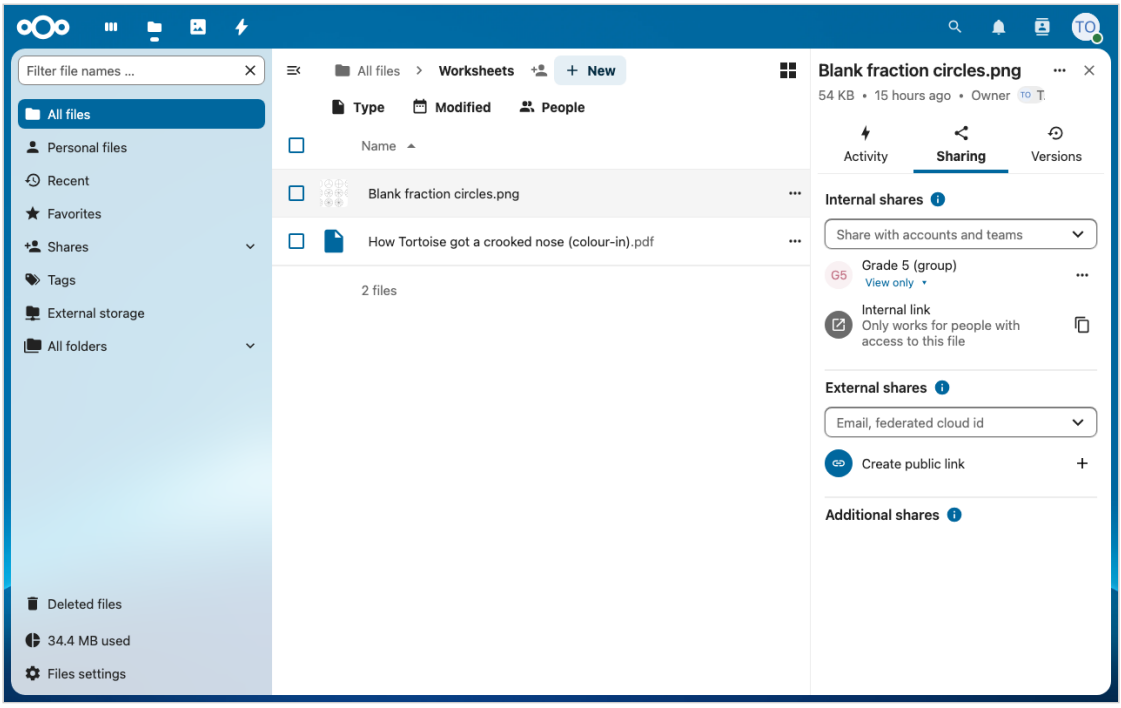


Figure 70. A file shared with Grade 5 (here a worksheet)

3. A learner in that group signs in and finds the file in **All files** and under **Recently shared**.

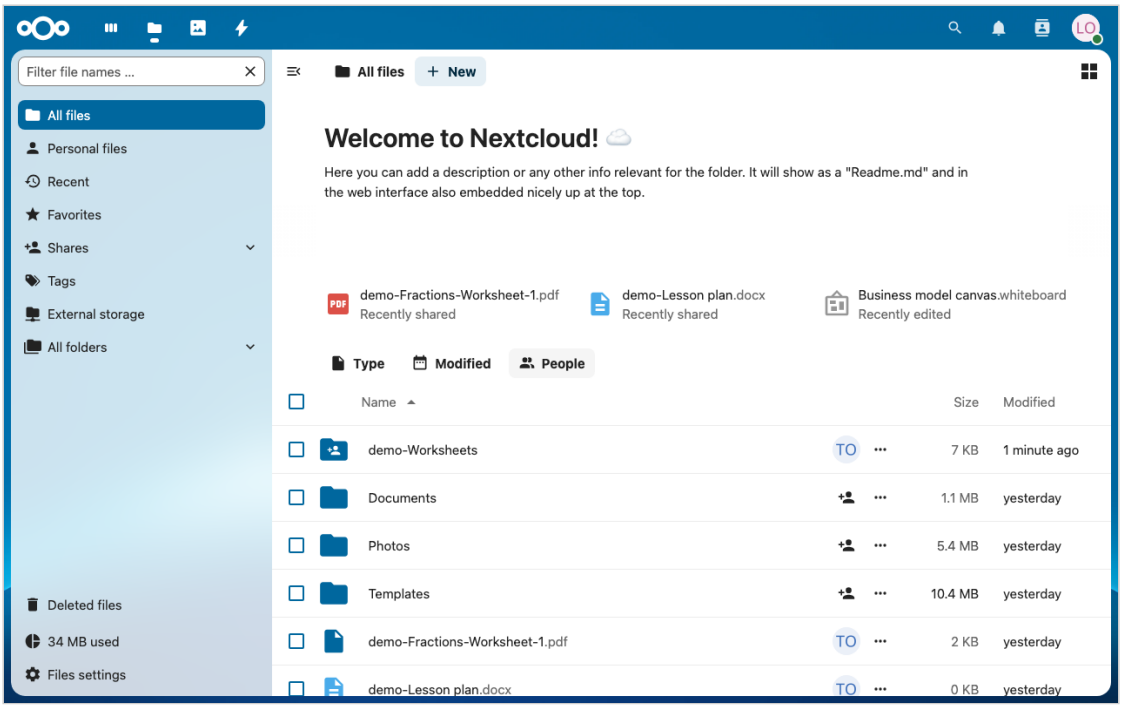


Figure 71. A learner sees the shared file

9. Keeping data safe: the Backup Disk

The operator looks after backups (IDEA for Operators: Tutorial). As a teacher you should know:

- A **Backup Disk** is a separate **IDEA Disk** with the **backup** badge in the Console.
- It saves **every app in full**: Kolibri (classes, learners, progress, and the channels), Nextcloud (accounts, and the files inside Nextcloud) and Wikipedia. On the kit that is about 164 GB per IDEA box, which fits on its 240 GB Backup Disk.
- It does **not** save the **School Files** folder (chapter 8).
- **During its backup, an app is stopped and can't be opened.** It starts again by itself when its backup is done. The first backup of all apps takes about 2 h 15 min, so backups are planned outside lesson time.
- How often backups run (on request, each time the disk is plugged in, or on a schedule) is set by your IDEA contact.
- If the IDEA box breaks, the operator restores the latest backup onto a spare disk. Only the work done after the last backup is lost.
- Never unplug a disk while a backup is running.

Appendix: demo accounts

This appendix lists the demo accounts and the demo content. The demo setup is on every IDEA box: the apps **Kolibri Demo** and **Nextcloud Demo**. Use them to follow the tutorial.

The demo school in Kolibri Demo is **IDEA Demo School**. The same usernames work in Kolibri Demo and in Nextcloud Demo.

Passwords:

- In Kolibri, learners have no password. They sign in with only their username.
- Every other demo password is the account name followed by `911!`, for example `teacher1911!`.

Username	Password	App	Role	Used in
admin	admin911!	Kolibri Demo	Kolibri admin of IDEA Demo School	3.2, 3.3
teacher1	teacher1911!	Kolibri Demo	Coach of class Grade 5	3
teacher2	teacher2911!	Kolibri Demo	Coach of class Grade 5	3
learner1	none (username only)	Kolibri Demo	Learner in Grade 5	3.5, 4
learner2	none (username only)	Kolibri Demo	Learner in Grade 5	4
learner3	none (username only)	Kolibri Demo	Learner in Grade 5	3.3, 3.5
admin	admin911!	Nextcloud Demo	Nextcloud administrator	
teacher1	teacher1911!	Nextcloud Demo	Teacher (group teachers)	5, 8
teacher2	teacher2911!	Nextcloud Demo	Teacher (group teachers)	5.2
learner1	learner1911!	Nextcloud Demo	Learner (group learners)	6, 8.2
learner2	learner2911!	Nextcloud Demo	Learner (group learners)	6.1
learner3	learner3911!	Nextcloud Demo	Learner (group learners)	

Demo content:

- Kolibri: the school **IDEA Demo School** with the class **Grade 5**.
- Kolibri Demo: three Kolibri channels, **Maths**, **Grade 5 Science** and **Early Reading** (videos, pictures and documents), with the exercises **Fractions practice** and **Times tables practice** (Maths) and **Reading: Rudo and the goat** (Early Reading).
- Kolibri Demo: the lessons **Maths lesson**, **Grade 5 Science lesson** and **Early Reading lesson** and the quiz **Grade 5 Practice Quiz**, for the class **Grade 5**. `learner3` has already done the quiz and **Fractions practice**, so reports show results. You make the **Fractions lesson** yourself in chapter 3.
- Nextcloud Demo: the folders **Worksheets** and **Teacher Notes** for `teacher1` and `teacher2`. You make the **Drop Zone** folder and the document **Lesson plan** yourself in chapter 5.
- Wikipedia (Demo): Wikipedia in simple English.

The school's own apps (not demo) start almost empty: **Kolibri** has only `admin` (the school creates its teachers and learners), **Nextcloud** has `admin`, `teacher1` and `learner1`. The passwords follow the same rule. Learners need a password in Nextcloud, not in Kolibri.

The pictures in the demo content are from Zimbabwe for now. The file `CREDITS.txt` in the demo content gives the source and licence of every file.